

# MATH6152GR

IZ TECÚM

ABSTRACT. Lecture notes for MATH6152GR.

## CONTENTS

|                                                                       |    |
|-----------------------------------------------------------------------|----|
| <b>Part 1. Analysis II<sup>1</sup></b>                                | 3  |
| 1. Preliminaries and Notation                                         | 3  |
| 1.1. Analytic Framework                                               | 3  |
| 2. The Laplacian                                                      | 4  |
| 2.1. Harmonic Functions                                               | 5  |
| 3. The Poisson Equation                                               | 7  |
| 3.1. Dirichlet Theory                                                 | 8  |
| 4. Potential Theory                                                   | 10 |
| 4.1. Singular Integrals                                               | 10 |
| 5. Schauder Estimates                                                 | 13 |
| 5.1. Variable Coefficients                                            | 13 |
| 6. Singular Integrals and $W^{2,p}$ Estimates                         | 15 |
| 6.1. Sobolev Estimates                                                | 15 |
| 7. Sobolev Spaces                                                     | 17 |
| 7.1. Core Tools                                                       | 18 |
| 8. The $H^1$ Theory                                                   | 19 |
| 8.1. Energy Methods                                                   | 19 |
| 9. De Giorgi - Nash - Moser theorem                                   | 21 |
| 9.1. Hölder Theory                                                    | 21 |
| 10. Minimal surface equation                                          | 22 |
| 10.1. Convex Graphs                                                   | 22 |
| 11. Boundary Problem                                                  | 23 |
| 11.1. Variational Problem                                             | 23 |
| <b>Part 2. Seminar talks<sup>2</sup></b>                              | 25 |
| 12. Spectral rigidity of random covers of compact hyperbolic surfaces | 25 |
| 12.1. Random Covers                                                   | 25 |
| <b>Part 3. Appendix</b>                                               | 28 |
| Appendix A. Foundational tools                                        | 28 |
| A.1. Core Estimates                                                   | 28 |

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<sup>1</sup>Based on lectures by Ovidiu Savin [2].

<sup>2</sup>Based on lectures by Elena Kim [1].



## Part 1. Analysis II<sup>3</sup>

### 1. PRELIMINARIES AND NOTATION

Throughout,  $\Omega \subset \mathbb{R}^n$  is open,  $C$  denotes a positive constant that may change between occurrences, and repeated upper and lower indices are summed. Derivatives are weak unless classical regularity is stated.

#### 1.1. Analytic Framework.

**Definition 1.1.1** (Multi-indices). For  $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$  and  $\beta \leq \alpha$ , set

$$|\alpha| = \sum_{i=1}^n \alpha_i, \quad \alpha! = \prod_{i=1}^n \alpha_i!, \quad D^\alpha = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}, \quad x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}, \quad \beta \leq \alpha \iff \beta_i \leq \alpha_i.$$

**Lemma 1.1.2** (Leibniz rule). For  $u, v \in C^\infty(\Omega)$  and every multi-index  $\alpha$ , one has

$$D^\alpha(uv) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v, \quad \binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!}.$$

**Definition 1.1.3** (Hölder spaces). For  $k \in \mathbb{Z}_{\geq 0}$  and  $\alpha \in (0, 1]$ , define

$$[u]_{C^{k,\alpha}(\Omega)} = \max_{|\beta|=k} \sup_{x \neq y} \frac{|D^\beta u(x) - D^\beta u(y)|}{|x-y|^\alpha}, \quad \|u\|_{C^{k,\alpha}(\Omega)} = \sum_{|\beta| \leq k} \|D^\beta u\|_{L^\infty(\Omega)} + [u]_{C^{k,\alpha}(\Omega)}.$$

The space  $C^{k,\alpha}(\Omega)$  consists of the functions with finite displayed norm.

**Proposition 1.1.4** (Hölder interpolation). Let  $\Omega$  be bounded and  $0 < \alpha < \beta \leq 1$ . For every  $\varepsilon > 0$ , one has

$$[u]_{C^{0,\alpha}(\Omega)} \leq \varepsilon [u]_{C^{0,\beta}(\Omega)} + C(\varepsilon, \alpha, \beta) \|u\|_{L^\infty(\Omega)}.$$

*Proof.* Set  $\delta = \varepsilon^{1/(\beta-\alpha)}$  and  $r = |x-y|$ . The two scales give

$$0 < r \leq \delta \implies \frac{|u(x) - u(y)|}{r^\alpha} \leq \delta^{\beta-\alpha} [u]_{C^{0,\beta}} = \varepsilon [u]_{C^{0,\beta}}, \quad r > \delta \implies \frac{|u(x) - u(y)|}{r^\alpha} \leq 2\delta^{-\alpha} \|u\|_{L^\infty}.$$

Taking the supremum proves the estimate.  $\square$

**Definition 1.1.5** (Sobolev spaces). For  $u, v \in L^1_{\text{loc}}(\Omega)$ , the weak derivative is determined by

$$(1.1.1) \quad D^\alpha u = v \iff \int_\Omega u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v \varphi \, dx \quad \forall \varphi \in C_c^\infty(\Omega).$$

For  $k \in \mathbb{Z}_{\geq 0}$  and  $1 \leq p < \infty$ , set

$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega), |\alpha| \leq k\}, \quad \|u\|_{W^{k,p}(\Omega)} = \left( \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p},$$

$$W_0^{k,p}(\Omega) = \overline{C_c^\infty(\Omega)}^{W^{k,p}}, \quad H^k(\Omega) = W^{k,2}(\Omega), \quad H_0^k(\Omega) = W_0^{k,2}(\Omega), \quad (u, v)_{H_0^1} = \int_\Omega \nabla u \cdot \nabla v \, dx.$$

$$W^{k,\infty}(\Omega) = \{u : D^\alpha u \in L^\infty(\Omega), |\alpha| \leq k\}, \quad \|u\|_{W^{k,\infty}(\Omega)} = \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}.$$

**Theorem 1.1.6** (Sobolev–Morrey). Let  $\Omega \subset \mathbb{R}^n$  be bounded, connected, and Lipschitz. If  $1 \leq p < n$  and  $p^* = np/(n-p)$ , then

$$W_0^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega), \quad \|u\|_{L^{p^*}(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}.$$

If  $p > n$  and  $\gamma = 1 - n/p$ , then every  $u \in W^{1,p}(\Omega)$  has a Hölder representative satisfying

$$W^{1,p}(\Omega) \hookrightarrow C^{0,\gamma}(\overline{\Omega}), \quad [u]_{C^{0,\gamma}(\overline{\Omega})} \leq C \|\nabla u\|_{L^p(\Omega)}.$$

<sup>3</sup>Based on lectures by Ovidiu Savin [2].

**Theorem 1.1.7** (Poincaré inequality). *Let  $\Omega \subset \mathbb{R}^n$  be bounded, connected, and Lipschitz, let  $1 \leq p < \infty$ , and write  $u_\Omega = \int_\Omega u dx$ . Then*

$$\|u - u_\Omega\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)} \quad \forall u \in W^{1,p}(\Omega), \quad \|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)} \quad \forall u \in W_0^{1,p}(\Omega).$$

**Theorem 1.1.8** (Rellich–Kondrachov). *Let  $\Omega \subset \mathbb{R}^n$  be bounded and Lipschitz. For  $1 \leq p < n$  and  $1 \leq q < p^*$ , one has*

$$W^{1,p}(\Omega) \Subset L^q(\Omega), \quad u_j \rightharpoonup u \text{ in } W^{1,p}(\Omega) \implies u_{j_k} \rightarrow u \text{ in } L^q(\Omega).$$

**Definition 1.1.9** (Elliptic operators). For a symmetric matrix  $A = (a^{ij})$ , the nondivergence and divergence forms are

$$Lu = a^{ij}u_{ij} + b^i u_i + cu, \quad \mathcal{L}u = \partial_i(a^{ij}u_j) + b^i u_i + cu.$$

They are uniformly elliptic when some  $0 < \lambda \leq \Lambda < \infty$  satisfy

$$(1.1.2) \quad \lambda |\mathbf{v}|^2 \leq a^{ij}(x)v_i v_j \leq \Lambda |\mathbf{v}|^2 \quad \forall (x, \mathbf{v}) \in \Omega \times \mathbb{R}^n.$$

**Theorem 1.1.10** (Maximum principle). *Let  $\Omega$  be bounded, let  $a^{ij}, b^i, c \in C(\overline{\Omega})$  satisfy (1.1.2), and suppose  $c \leq 0$ . For  $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ , one has*

$$Lu \geq Lv \text{ in } \Omega, \quad u \leq v \text{ on } \partial\Omega \implies u \leq v \text{ on } \overline{\Omega}.$$

*Proof.* Set  $w = u - v$  and make the differential inequality strict with  $\psi(x) = e^{\kappa x_1}$ . For  $\kappa$  sufficiently large,

$$L\psi = e^{\kappa x_1}(a^{11}\kappa^2 + b^1\kappa + c) \geq e^{\kappa x_1}(\lambda\kappa^2 - \|b^1\|_{L^\infty}\kappa - \|c\|_{L^\infty}) > 0, \quad L(w + \varepsilon\psi) > 0.$$

At a positive interior maximum of  $w_\varepsilon = w + \varepsilon\psi$ ,

$$Dw_\varepsilon = 0, \quad D^2w_\varepsilon \leq 0 \xrightarrow{1.1.9} Lw_\varepsilon \leq cw_\varepsilon \leq 0, \quad Lw_\varepsilon > 0,$$

a contradiction. Hence

$$w_\varepsilon(x) \leq \max\{0, \max_{\partial\Omega} w_\varepsilon\} \leq \varepsilon \max_{\partial\Omega} \psi, \quad \varepsilon \downarrow 0 \implies w \leq 0.$$

This proves the comparison principle.  $\square$

**Theorem 1.1.11** (Schauder estimate). *Let  $L$  satisfy (1.1.2) on  $B_1$ , let  $a^{ij}, b^i, c \in C^{0,\alpha}(B_1)$ , and suppose  $Lu = f \in C^{0,\alpha}(B_1)$ . Then*

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}), \quad C = C(n, \alpha, \lambda, \Lambda, \|a\|_{C^{0,\alpha}}, \|b\|_{C^{0,\alpha}}, \|c\|_{C^{0,\alpha}}).$$

**Theorem 1.1.12** (Calderón–Zygmund estimate). *Let  $1 < p < \infty$ , let  $L$  satisfy (1.1.2) on  $B_1$ , and assume its principal coefficients are continuous and its lower-order coefficients are bounded. For  $u \in W^{2,p}(B_1)$  and  $Lu = f \in L^p(B_1)$ , one has*

$$\|u\|_{W^{2,p}(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}), \quad C = C(n, p, \lambda, \Lambda, \omega_A, \|b\|_{L^\infty}, \|c\|_{L^\infty}).$$

## 2. THE LAPLACIAN

The Laplacian is the constant-coefficient model for the elliptic operators of Definition 1.1.9. Throughout this section,  $n \geq 2$ ,  $\omega_n = |B_1|$ , and the notation of Section 1 remains in force.

### 2.1. Harmonic Functions.

**Definition 2.1.1** (Laplacian). For  $u \in C^2(\Omega)$ , define

$$\Delta u = \sum_{i=1}^n \partial_i^2 u = \operatorname{div}(\nabla u), \quad u \text{ harmonic} \iff \Delta u = 0, \quad u \text{ subharmonic} \iff \Delta u \geq 0.$$

For  $u \in L^1_{\text{loc}}(\Omega)$ , these inequalities are distributional with

$$\langle \Delta u, \varphi \rangle = \int_{\Omega} u \Delta \varphi dx \quad \forall \varphi \in C_c^\infty(\Omega).$$

**Theorem 2.1.2** (Green identities). Let  $\Omega \subset \mathbb{R}^n$  be bounded with  $C^1$  boundary and let  $u, v \in C^2(\Omega) \cap C^1(\overline{\Omega})$ . With  $\nu$  the outward unit normal, one has

$$\begin{aligned} \int_{\Omega} \Delta u dx &= \int_{\partial\Omega} \partial_\nu u d\sigma, & \int_{\Omega} \nabla u \cdot \nabla v dx &= - \int_{\Omega} u \Delta v dx + \int_{\partial\Omega} u \partial_\nu v d\sigma, \\ \int_{\Omega} (v \Delta u - u \Delta v) dx &= \int_{\partial\Omega} (v \partial_\nu u - u \partial_\nu v) d\sigma. \end{aligned}$$

**Proposition 2.1.3** (Spherical means). For  $u \in C^2(\Omega)$  and  $B_r(x_0) \Subset \Omega$ , set

$$M_\partial(r) = \oint_{\partial B_r(x_0)} u d\sigma, \quad M_B(r) = \int_{B_r(x_0)} u dx.$$

As  $r \downarrow 0$ , one has

$$\begin{aligned} M'_\partial(r) &= \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x_0)} \Delta u dx, & M_\partial(r) &= u(x_0) + \frac{r^2}{2n} \Delta u(x_0) + o(r^2), \\ M_B(r) &= u(x_0) + \frac{r^2}{2(n+2)} \Delta u(x_0) + o(r^2). \end{aligned}$$

**Theorem 2.1.4** (Mean value property). For  $u \in C^2(\Omega)$ , harmonicity is equivalent to both mean value identities

$$\Delta u = 0 \iff u(x_0) = \oint_{\partial B_r(x_0)} u d\sigma \iff u(x_0) = \int_{B_r(x_0)} u dx \quad \forall B_r(x_0) \Subset \Omega.$$

*Proof.* Differentiate the spherical mean and integrate in polar coordinates. Proposition 2.1.3 gives

$$\Delta u = 0 \stackrel{2.1.3}{\iff} M'_\partial(r) = 0 \iff M_\partial(r) = u(x_0), \quad M_B(r) = \frac{n}{r^n} \int_0^r s^{n-1} M_\partial(s) ds = u(x_0).$$

Conversely, either mean value identity and the corresponding expansion in Proposition 2.1.3 forces  $\Delta u(x_0) = 0$ . This proves all three equivalences.  $\square$

**Theorem 2.1.5** (Fundamental solution). The locally integrable function

$$(2.1.1) \quad \Gamma(x) = \begin{cases} \frac{|x|^{2-n}}{n(n-2)\omega_n}, & n \geq 3, \\ -\frac{1}{2\pi} \log|x|, & n = 2, \end{cases} \quad \Delta \Gamma = -\delta_0 \text{ in } \mathcal{D}'(\mathbb{R}^n).$$

*Proof.* Radial differentiation gives the classical equation and the unit inward flux at the pole.

$$r > 0 \implies \Delta \Gamma = \Gamma''(r) + \frac{n-1}{r} \Gamma'(r) = 0, \quad \int_{\partial B_r} \partial_r \Gamma d\sigma = -1.$$

For  $\varphi \in C_c^\infty(\mathbb{R}^n)$ , Green's identity on  $\mathbb{R}^n \setminus B_\varepsilon$  yields

$$\langle \Delta \Gamma, \varphi \rangle = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n \setminus B_\varepsilon} \Gamma \Delta \varphi dx = \lim_{\varepsilon \downarrow 0} \int_{\partial B_\varepsilon} (-\Gamma \partial_r \varphi + \varphi \partial_r \Gamma) d\sigma = -\varphi(0).$$

Thus (2.1.1) holds distributionally.  $\square$

**Definition 2.1.6** (Regularization). For  $\rho > 0$ , replace the pole in (2.1.1) by the unique radial quadratic matching  $\Gamma$  to first order on  $\partial B_\rho$ . The resulting function satisfies

$$\Gamma_\rho(x) = \begin{cases} \Gamma(x), & |x| \geq \rho, \\ a_\rho + b_\rho |x|^2, & |x| < \rho, \end{cases} \quad \Gamma_\rho \in C^1(\mathbb{R}^n), \quad \Delta \Gamma_\rho = -\frac{\chi_{B_\rho}}{|B_\rho|}.$$

Moreover,  $\Gamma_\rho \rightarrow \Gamma$  in  $L^p_{\text{loc}}$  for  $p < n/(n-2)$  when  $n \geq 3$  and for every finite  $p$  when  $n = 2$ .

**Proposition 2.1.7** (Green representation). Let  $\Omega$  be bounded with  $C^1$  boundary, let  $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ , and let  $x_0 \in \Omega$ . Then

$$(2.1.2) \quad u(x_0) = - \int_{\Omega} \Gamma(x-x_0) \Delta u(x) dx + \int_{\partial\Omega} (\Gamma(x-x_0) \partial_\nu u(x) - u(x) \partial_\nu \Gamma(x-x_0)) d\sigma(x).$$

**Theorem 2.1.8** (Weyl lemma). Let  $u \in L^1_{\text{loc}}(\Omega)$  satisfy  $\Delta u = 0$  distributionally. For every  $B_R(x_0) \Subset \Omega$  and multi-index  $\alpha$ , one has

$$u \in C^\infty(\Omega), \quad \|D^\alpha u\|_{L^\infty(B_{R/2}(x_0))} \leq \frac{C_{n,|\alpha|}}{R^{n+|\alpha|}} \|u\|_{L^1(B_R(x_0))}.$$

**Corollary 2.1.9** (Analyticity). Every harmonic function is real analytic, namely

$$\Delta u = 0 \text{ in } \Omega \implies u \in C^\omega(\Omega).$$

**Theorem 2.1.10** (Liouville theorem). Let  $u$  be harmonic on  $\mathbb{R}^n$  and satisfy  $|u(x)| \leq C(1+|x|^m)$  for some  $m \in \mathbb{Z}_{\geq 0}$ . Then

$u$  is a harmonic polynomial of degree at most  $m$ .

*Proof.* Fix  $x_0 \in \mathbb{R}^n$  and  $|\alpha| = k > m$ . Rescaling Theorem 2.1.8 gives

$$|D^\alpha u(x_0)| \leq \frac{C_{n,k}}{R^{n+k}} \|u\|_{L^1(B_R(x_0))} \leq C_{n,k} R^{m-k} \xrightarrow{R \rightarrow \infty} 0, \quad D^\alpha u \equiv 0.$$

All derivatives of order greater than  $m$  vanish, proving the claim.  $\square$

**Theorem 2.1.11** (Subharmonicity). Let  $u \in L^1_{\text{loc}}(\Omega)$  have an upper-semicontinuous representative. The following conditions are equivalent:

$$\begin{aligned} \Delta u \geq 0 &\iff u(x_0) \leq \int_{\partial B_r(x_0)} u d\sigma \iff u(x_0) \leq \int_{B_r(x_0)} u dx \quad \forall B_r(x_0) \Subset \Omega, \\ \Delta u \geq 0 &\iff (\phi \in C^2, \phi \geq u \text{ near } x_0, \phi(x_0) = u(x_0)) \implies \Delta \phi(x_0) \geq 0. \end{aligned}$$

**Proposition 2.1.12** (Harmonic replacement). Let  $u$  be continuous and subharmonic, and let  $h$  be harmonic on  $B_r(x_0) \Subset \Omega$  with  $h = u$  on  $\partial B_r(x_0)$ . The replacement

$$\tilde{u} = \begin{cases} h, & B_r(x_0), \\ u, & \Omega \setminus B_r(x_0), \end{cases} \quad \tilde{u} \geq u, \quad \Delta \tilde{u} \geq 0$$

is subharmonic on  $\Omega$ .

**Theorem 2.1.13** (Weak maximum principle). Let  $\Omega$  be bounded and  $u \in C^2(\Omega) \cap C(\overline{\Omega})$ . The maximum principle of Theorem 1.1.10 gives

$$\Delta u \geq 0 \implies \max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \quad \Delta u \leq 0 \implies \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

**Theorem 2.1.14** (Strong maximum principle). Let  $\Omega$  be connected and let  $u \in C^2(\Omega)$  satisfy  $\Delta u \geq 0$ . If  $u$  attains an interior maximum, then

$$u \equiv \max_{\overline{\Omega}} u.$$

*Proof.* Let  $M = \max_{\Omega} u$  and  $A = \{x \in \Omega : u(x) = M\}$ . For  $x_0 \in A$  and  $B_r(x_0) \Subset \Omega$ , Theorem 2.1.11 gives

$$M = u(x_0) \leq \int_{B_r(x_0)} u dx \leq M \implies u \equiv M \text{ on } B_r(x_0) \implies A \text{ is open and closed in } \Omega.$$

Connectedness gives  $A = \Omega$ , proving the result.  $\square$

**Theorem 2.1.15** (Hopf lemma). *Let  $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$  satisfy  $\Delta u \geq 0$ , and suppose  $u$  has a strict maximum at  $x_0 \in \partial\Omega$  where  $\Omega$  has an interior tangent ball. For the outward unit normal  $v$ , one has*

$$u(x_0) > u(x) \quad \forall x \in \Omega \implies \partial_v u(x_0) > 0.$$

**Corollary 2.1.16** (Dirichlet uniqueness). *Let  $\Omega$  be bounded and connected. For  $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ , one has*

$$\Delta u = \Delta v \text{ in } \Omega, \quad u = v \text{ on } \partial\Omega \xrightarrow{2.1.13} u - v \leq 0 \wedge v - u \leq 0 \implies u \equiv v.$$

**Theorem 2.1.17** (Derivative estimates). *Let  $u$  be harmonic in  $B_R(x_0)$ . For every multi-index  $\alpha$ , one has*

$$(2.1.3) \quad |\nabla u(x_0)| \leq \frac{n}{R} \|u\|_{L^\infty(B_R(x_0))}, \quad |D^\alpha u(x_0)| \leq \frac{C_n^{|\alpha|+1} |\alpha|!}{R^{|\alpha|}} \|u\|_{L^\infty(B_R(x_0))}.$$

**Theorem 2.1.18** (Harnack inequality). *Let  $u \geq 0$  be harmonic in  $B_R(x_0)$ . Then*

$$\sup_{B_{R/2}(x_0)} u \leq C_n \inf_{B_{R/2}(x_0)} u.$$

*Proof.* The mean value property first gives a local comparison. If  $B_{2r}(z) \subset B_R(x_0)$  and  $y \in B_r(z)$ , then

$$u(y) = \int_{B_r(y)} u dx \leq \frac{|B_{2r}|}{|B_r|} \int_{B_{2r}(z)} u dx = 2^n u(z).$$

Any two points  $x, y \in B_{R/2}(x_0)$  are joined by at most eight such steps with  $r = R/8$ . Hence

$$u(x) \leq 2^{8n} u(y), \quad x, y \in B_{R/2}(x_0) \implies \sup_{B_{R/2}(x_0)} u \leq 2^{8n} \inf_{B_{R/2}(x_0)} u.$$

This proves the estimate with  $C_n = 2^{8n}$ .  $\square$

**Corollary 2.1.19** (Nonnegative Liouville theorem). *Every nonnegative entire harmonic function is constant, namely*

$$u \geq 0, \quad \Delta u = 0 \text{ in } \mathbb{R}^n \implies u \equiv u(0).$$

**Theorem 2.1.20** (Hölder estimate). *Let  $u$  be harmonic in  $B_1$ . For some  $\alpha = \alpha(n) \in (0, 1)$ , one has*

$$[u]_{C^{0,\alpha}(B_{1/2})} \leq C_n \|u\|_{L^\infty(B_1)}.$$

*Proof.* Write

$$M(r) = \sup_{B_r} u, \quad m(r) = \inf_{B_r} u, \quad \eta(r) = M(r) - m(r).$$

Apply Theorem 2.1.18 to the nonnegative harmonic functions  $M(r) - u$  and  $u - m(r)$ :

$$M(r) - m(r/2) \leq C_H (M(r) - M(r/2)), \quad M(r/2) - m(r) \leq C_H (m(r/2) - m(r)).$$

Adding and iterating gives

$$\eta(r/2) \leq \theta \eta(r), \quad \theta = \frac{C_H - 1}{C_H + 1} \in (0, 1), \quad \eta(2^{-k}) \leq \theta^k \eta(1) = 2^{-k\alpha} \eta(1), \quad \alpha = -\log_2 \theta.$$

Scaling this oscillation decay on balls contained in  $B_1$  proves the Hölder estimate.  $\square$

### 3. THE POISSON EQUATION

The Dirichlet problem is reduced to harmonic extension by Green functions, Poisson kernels, barriers, and Newtonian potentials.

### 3.1. Dirichlet Theory.

**Definition 3.1.1** (Dirichlet problem). Let  $\Omega \subset \mathbb{R}^n$  be bounded,  $f \in C(\Omega)$ , and  $\varphi \in C(\partial\Omega)$ . The Dirichlet problem is

$$\begin{cases} \Delta u = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega, \end{cases} \quad u \in C^2(\Omega) \cap C(\overline{\Omega}).$$

**Theorem 3.1.2** (Uniqueness). For  $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ , one has

$$\begin{cases} \Delta u = \Delta v & \text{in } \Omega, \\ u = v & \text{on } \partial\Omega \end{cases} \xrightarrow{2.1.16} u \equiv v.$$

**Definition 3.1.3** (Green function). Assume every continuous boundary datum on the bounded connected domain  $\Omega$  has a harmonic extension. For  $y \in \Omega$ , let  $h_y$  extend  $\Gamma(\cdot - y)|_{\partial\Omega}$  and define

$$G_\Omega(x, y) = \Gamma(x - y) - h_y(x), \quad \begin{cases} \Delta_x G_\Omega(x, y) = -\delta_y & \text{in } \Omega, \\ G_\Omega(x, y) = 0 & \text{on } \partial\Omega. \end{cases}$$

**Theorem 3.1.4** (Green representation). Let  $\Omega$  be a bounded connected  $C^2$  domain admitting a Green function, and let  $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ . With  $f = \Delta u$ ,  $\varphi = u|_{\partial\Omega}$ , and

$$P_\Omega(y, x) = -\partial_{v_x} G_\Omega(x, y),$$

one has

$$(3.1.1) \quad u(y) = - \int_{\Omega} G_\Omega(x, y) f(x) dx + \int_{\partial\Omega} P_\Omega(y, x) \varphi(x) d\sigma(x).$$

*Proof.* Write  $\Gamma_y = \Gamma(\cdot - y)$ . Green's identity and  $h_y = \Gamma_y$  on  $\partial\Omega$  give

$$\int_{\Omega} h_y f dx = \int_{\partial\Omega} (h_y \partial_\nu u - u \partial_\nu h_y) d\sigma = \int_{\partial\Omega} (\Gamma_y \partial_\nu u - \varphi \partial_\nu \Gamma_y) d\sigma.$$

Consequently,

$$u(y) \xrightarrow{2.1.7} - \int_{\Omega} f \Gamma_y dx + \int_{\partial\Omega} (\Gamma_y \partial_\nu u - \varphi \partial_\nu \Gamma_y) d\sigma = - \int_{\Omega} f (\Gamma_y - h_y) dx - \int_{\partial\Omega} \varphi \partial_\nu (\Gamma_y - h_y) d\sigma,$$

which is 3.1.1.  $\square$

**Proposition 3.1.5** (Green properties). For every bounded connected domain admitting a Green function,

$$x \neq y \implies G_\Omega(x, y) = G_\Omega(y, x) > 0.$$

If  $\Omega_1 \subset \Omega_2$  admit Green functions, then

$$x, y \in \Omega_1, \quad x \neq y \xrightarrow{2.1.13} 0 < G_{\Omega_1}(x, y) \leq G_{\Omega_2}(x, y).$$

**Proposition 3.1.6** (Model Green functions). Let  $\mathbb{H} = \{x_n > 0\}$  and  $y^\# = (y', -y_n)$ . For  $x, y \in \mathbb{H}$ ,

$$G_{\mathbb{H}}(x, y) = \Gamma(x - y) - \Gamma(x - y^\#).$$

For  $x, y \in B_1$  and  $y \neq 0$ , let  $y^* = y/|y|^2$ . Then

$$G_{B_1}(x, y) = \Gamma(x - y) - \Gamma(|y|(x - y^*)), \quad G_{B_1}(x, 0) = \Gamma(x) - \Gamma(1).$$

Their Poisson kernels are

$$(3.1.2) \quad P_{\mathbb{H}}(x, \zeta) = \frac{2x_n}{n\omega_n|x - \zeta|^n}, \quad P_{B_1}(x, \zeta) = \frac{1 - |x|^2}{n\omega_n|x - \zeta|^n}, \quad \int_{\partial B_1} P_{B_1}(x, \zeta) d\sigma(\zeta) = 1.$$



**Theorem 3.1.7** (Poisson formula). *For  $\varphi \in C(\partial B_1)$ , define*

$$\mathcal{P}\varphi(x) = \int_{\partial B_1} P_{B_1}(x, \zeta) \varphi(\zeta) d\sigma(\zeta).$$

*Then  $\mathcal{P}\varphi$  is the unique harmonic extension and*

$$\mathcal{P}\varphi \in C^\infty(B_1) \cap C(\overline{B_1}), \quad \begin{cases} \Delta \mathcal{P}\varphi = 0 & \text{in } B_1, \\ \mathcal{P}\varphi = \varphi & \text{on } \partial B_1, \end{cases} \quad \|\mathcal{P}\varphi\|_{L^\infty(B_1)} \leq \|\varphi\|_{L^\infty(\partial B_1)}.$$

*Proof.* The kernel in 3.1.2 is positive, harmonic in  $x$ , and has mass one. Fix  $\zeta_0 \in \partial B_1$  and choose  $\delta > 0$  such that  $|\varphi(\zeta) - \varphi(\zeta_0)| < \varepsilon$  for  $|\zeta - \zeta_0| < \delta$ . Then

$$|\mathcal{P}\varphi(x) - \varphi(\zeta_0)| \leq \varepsilon + 2\|\varphi\|_{L^\infty} \int_{|\zeta - \zeta_0| \geq \delta} P_{B_1}(x, \zeta) d\sigma(\zeta).$$

For  $|x - \zeta_0| < \delta/2$ ,

$$|\zeta - \zeta_0| \geq \delta \implies |x - \zeta| \geq \frac{\delta}{2}, \quad P_{B_1}(x, \zeta) \leq C_{n,\delta} (1 - |x|^2) \xrightarrow{x \rightarrow \zeta_0} 0.$$

Thus  $\mathcal{P}\varphi(x) \rightarrow \varphi(\zeta_0)$ ; harmonicity and the norm bound follow from differentiation and positivity.  $\square$

**Definition 3.1.8** (Barriers). A barrier at  $\zeta \in \partial\Omega$  is a function  $b \in C^2(\Omega) \cap C(\overline{\Omega})$  satisfying

$$\begin{cases} \Delta b \leq 0 & \text{in } \Omega, \\ b > 0 & \text{on } \overline{\Omega} \setminus \{\zeta\}, \\ b(\zeta) = 0. \end{cases}$$

The exterior ball condition at  $\zeta$  means that some ball satisfies

$$B_r(y) \subset \mathbb{R}^n \setminus \Omega, \quad \overline{B_r(y)} \cap \overline{\Omega} = \{\zeta\}.$$

**Proposition 3.1.9** (Exterior barriers). *Under the exterior ball condition, a barrier at  $\zeta$  is*

$$b_\zeta(x) = \begin{cases} r^{2-n} - |x - y|^{2-n}, & n \geq 3, \\ \log \frac{|x - y|}{r}, & n = 2. \end{cases}$$

*Proof.* The exterior tangency and the fundamental solution give

$$x \in \overline{\Omega} \setminus \{\zeta\} \implies |x - y| > r \implies b_\zeta(x) > 0, \quad |\zeta - y| = r \implies b_\zeta(\zeta) = 0, \quad y \notin \overline{\Omega} \xrightarrow{2.1.1} \Delta b_\zeta = 0.$$

Hence  $b_\zeta$  is a barrier.  $\square$

**Definition 3.1.10** (Perron envelope). For  $\varphi \in C(\partial\Omega)$ , let

$$\mathcal{S}_\varphi = \left\{ v : v \text{ is subharmonic in } \Omega, \limsup_{x \rightarrow \zeta} v(x) \leq \varphi(\zeta) \forall \zeta \in \partial\Omega \right\}, \quad H_\varphi(x) = \sup_{v \in \mathcal{S}_\varphi} v(x).$$

**Theorem 3.1.11** (Perron method). *Let  $\Omega$  be bounded and connected and let  $\varphi \in C(\partial\Omega)$ . Then*

$$\Delta H_\varphi = 0 \text{ in } \Omega, \quad \zeta \text{ admits a barrier} \implies \lim_{x \rightarrow \zeta} H_\varphi(x) = \varphi(\zeta).$$

*Consequently,*

$$\Omega \text{ satisfies the exterior ball condition on } \partial\Omega \xrightarrow{3.1.9} H_\varphi \in C(\overline{\Omega}), \quad \begin{cases} \Delta H_\varphi = 0 & \text{in } \Omega, \\ H_\varphi = \varphi & \text{on } \partial\Omega. \end{cases}$$

**Definition 3.1.12** (Newtonian potential). For compactly supported  $f \in L^p(\mathbb{R}^n)$  with  $p > n/2$ , define

$$(3.1.3) \quad \mathcal{N}f(x) = (\Gamma * f)(x) = \int_{\mathbb{R}^n} \Gamma(x-y)f(y)dy, \quad -\Delta \mathcal{N}f = f \text{ in } \mathcal{D}'(\mathbb{R}^n).$$

**Theorem 3.1.13** (Potential regularity). Let  $\Omega$  be bounded and extend  $f$  by zero outside  $\Omega$ . Then

$$p > \frac{n}{2}, \quad f \in L^p(\Omega) \implies \mathcal{N}f \in C(\mathbb{R}^n),$$

while

$$f \in L^\infty(\Omega) \implies \mathcal{N}f \in C^1(\mathbb{R}^n), \quad \partial_i \mathcal{N}f = f * \partial_i \Gamma.$$

**Theorem 3.1.14** ( $L^p$  solvability). Let  $\Omega$  satisfy the exterior ball condition, let  $f \in L^p(\Omega)$  with  $p > n/2$ , and let  $\varphi \in C(\partial\Omega)$ . Then the unique distributional solution of Definition 3.1.1 is constructed by

$$h = H_{\varphi + \mathcal{N}f|_{\partial\Omega}}, \quad u = h - \mathcal{N}f, \quad \xrightarrow{3.1.3} \begin{cases} \Delta u = 0 - \Delta \mathcal{N}f = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

Existence follows from Theorem 3.1.11 and uniqueness from Theorem 3.1.2.

#### 4. POTENTIAL THEORY

The identity 3.1.3 reduces constant-coefficient elliptic regularity to estimates for the singular Hessian of  $\Gamma$ .

##### 4.1. Singular Integrals.

**Theorem 4.1.1** (Newtonian regularity). Let  $0 < \alpha < 1$ ,  $f \in C_c^{0,\alpha}(\mathbb{R}^n)$ , and  $w = \mathcal{N}f$ . For  $z \neq 0$ , set

$$K_{ij}(z) = D_{ij}\Gamma(z) = \frac{1}{n\omega_n} \left( \frac{nz_i z_j}{|z|^{n+2}} - \frac{\delta_{ij}}{|z|^n} \right), \quad \int_{\partial B_r} K_{ij} d\sigma = 0.$$

Then  $w \in C^2(\mathbb{R}^n)$  and

$$(4.1.1) \quad D_{ij}w(x) = -\frac{\delta_{ij}}{n}f(x) + T_{ij}f(x), \quad T_{ij}f(x) = \text{p.v.} \int_{\mathbb{R}^n} K_{ij}(z)f(x-z)dz.$$

Equivalently,

$$T_{ij}f(x) = \int_{B_1} (f(x-z) - f(x))K_{ij}(z)dz + \int_{\mathbb{R}^n \setminus B_1} f(x-z)K_{ij}(z)dz.$$

*Proof.* Regularize  $\Gamma$  as in Definition 2.1.6 and set  $w_\rho = f * \Gamma_\rho$ . Differentiation gives

$$D_{ij}w_\rho(x) = -\frac{\delta_{ij}}{n} \int_{B_\rho} f(x-z)dz + \int_{|z|>\rho} f(x-z)K_{ij}(z)dz.$$

Spherical cancellation removes the constant part near the pole, while

$$\int_{\varepsilon < |z| < r} |f(x-z) - f(x)| |K_{ij}(z)| dz \leq C[f]_{C^{0,\alpha}} \int_\varepsilon^r s^{\alpha-1} ds \leq C[f]_{C^{0,\alpha}} r^\alpha.$$

The far integral is smooth because  $f$  is compactly supported. Hence

$$w_\rho \rightarrow w \text{ in } C_{\text{loc}}^1, \quad D_{ij}w_\rho \rightarrow -\frac{\delta_{ij}}{n}f + T_{ij}f \text{ locally uniformly} \implies w \in C^2(\mathbb{R}^n),$$

which proves 4.1.1. □

**Definition 4.1.2** (Singular kernels). A Calderón–Zygmund kernel is a function  $K \in C^1(\mathbb{R}^n \setminus \{0\})$  satisfying

$$K(tz) = t^{-n}K(z), \quad \int_{\partial B_1} K d\sigma = 0, \quad |K(z)| + |z||\nabla K(z)| \leq A_K |z|^{-n}.$$

Its truncated and principal-value operators are

$$T_\varepsilon f(x) = \int_{|x-y|>\varepsilon} K(x-y)f(y) dy, \quad Tf(x) = \lim_{\varepsilon \downarrow 0} T_\varepsilon f(x).$$

The kernels  $K_{ij}$  in Theorem 4.1.1 satisfy these conditions.

**Theorem 4.1.3** (Hölder boundedness). *Let  $K$  satisfy Definition 4.1.2. For  $0 < \alpha < 1$  and  $f \in C_c^{0,\alpha}(\mathbb{R}^n)$ , one has*

(4.1.2)

$$\|Tf\|_{C^{0,\alpha}(\mathbb{R}^n)} \leq C(n, \alpha) A_K [f]_{C^{0,\alpha}(\mathbb{R}^n)}, \quad \text{Supp}(f) \subset B_R \implies \|Tf\|_{C^{0,\alpha}(\mathbb{R}^n)} \leq C(n, \alpha, A_K, R) \|f\|_{C^{0,\alpha}(\mathbb{R}^n)}.$$

*Proof.* Fix  $x, y \in \mathbb{R}^n$  and set  $d = |x - y|$ . Cancellation and translation split the difference into near and far terms satisfying

$$|I_{\text{near}}| \leq C A_K [f]_{C^{0,\alpha}} \int_0^{4d} r^{\alpha-1} dr \leq C A_K [f]_{C^{0,\alpha}} d^\alpha, \\ |I_{\text{far}}| \leq C A_K [f]_{C^{0,\alpha}} d \int_{3d}^\infty r^{\alpha-2} dr \leq C A_K [f]_{C^{0,\alpha}} d^\alpha.$$

Thus  $|Tf(x) - Tf(y)| \leq C A_K [f]_{C^{0,\alpha}} |x - y|^\alpha$ ; the support condition and the same near–far splitting control  $\|Tf\|_{L^\infty}$ .  $\square$

**Lemma 4.1.4** (Fourier multiplier). *The principal-value Fourier transform of every kernel in Definition 4.1.2 is a bounded multiplier  $m$  homogeneous of degree zero. For the kernels  $K_{ij}$ ,*

$$\widehat{T_{ij}f}(\xi) = m_{ij}(\xi) \widehat{f}(\xi), \quad m_{ij}(\xi) = \frac{\delta_{ij}}{n} - \frac{\xi_i \xi_j}{|\xi|^2}, \quad \|m\|_{L^\infty} \leq C(n) A_K.$$

**Corollary 4.1.5** ( $L^2$  boundedness). *Every Calderón–Zygmund operator extends uniquely to  $L^2(\mathbb{R}^n)$  and satisfies*

$$\|Tf\|_{L^2(\mathbb{R}^n)} \leq C(n) A_K \|f\|_{L^2(\mathbb{R}^n)}.$$

*Proof.* For  $f \in C_c^\infty(\mathbb{R}^n)$ , Plancherel and Lemma 4.1.4 give

$$\widehat{Tf} = m \widehat{f} \xrightarrow{4.1.4} \|Tf\|_{L^2}^2 = \int_{\mathbb{R}^n} |m(\xi)|^2 |\widehat{f}(\xi)|^2 d\xi \leq C(n) A_K^2 \|f\|_{L^2}^2.$$

Density gives the extension.  $\square$

**Lemma 4.1.6** (Calderón–Zygmund decomposition). *Let  $f \in L^1(\mathbb{R}^n)$  and  $\lambda > 0$ . There exist disjoint dyadic cubes  $Q_k$  and a decomposition  $f = g + \sum_k b_k$  satisfying*

$$|g| \leq 2^n \lambda \text{ a.e.}, \quad \text{Supp}(b_k) \subset Q_k, \quad \int_{Q_k} b_k dx = 0, \quad \lambda < \int_{Q_k} |f| dx \leq 2^n \lambda, \\ \sum_k |Q_k| \leq \frac{\|f\|_{L^1}}{\lambda}, \quad \|g\|_{L^1} \leq \|f\|_{L^1}, \quad \sum_k \|b_k\|_{L^1} \leq 2 \|f\|_{L^1}.$$

**Theorem 4.1.7** (Weak type). *Every Calderón–Zygmund operator satisfies*

$$|\{x \in \mathbb{R}^n : |Tf(x)| > \lambda\}| \leq \frac{C(n) A_K}{\lambda} \|f\|_{L^1(\mathbb{R}^n)} \quad \forall f \in L^1(\mathbb{R}^n), \quad \lambda > 0.$$

**Theorem 4.1.8** (Marcinkiewicz interpolation). *Let  $T$  be sublinear and of weak types  $(p_0, p_0)$  and  $(p_1, p_1)$ , where  $1 \leq p_0 < p_1 \leq \infty$ . Then*

$$p_0 < p < p_1 \implies \|Tf\|_{L^p} \leq C(p_0, p, p_1) \|f\|_{L^p}.$$

**Theorem 4.1.9** (Calderón–Zygmund theorem). *Let  $K$  satisfy Definition 4.1.2. Then*

$$T : L^1 \rightarrow L^{1,\infty}, \quad T : L^2 \rightarrow L^2 \xrightarrow{4.1.8} T : L^p \rightarrow L^p \quad (1 < p \leq 2),$$

*and duality applied to the adjoint kernel  $K(-z)$  gives*

$$1 < p < \infty \implies \|Tf\|_{L^p(\mathbb{R}^n)} \leq C(n, p) A_K \|f\|_{L^p(\mathbb{R}^n)}.$$

*Consequently,*

$$4.1.1 \implies \|D^2 \mathcal{N}f\|_{L^p(\mathbb{R}^n)} \leq C(n, p) \|f\|_{L^p(\mathbb{R}^n)}.$$

**Theorem 4.1.10** (Interior Schauder estimate). *Let  $0 < \alpha < 1$ ,  $u \in C^2(B_1)$ , and  $\Delta u = f \in C^{0,\alpha}(B_1)$ . Then*

$$(4.1.3) \quad \|u\|_{C^{2,\alpha}(B_{1/2})} \leq C(n, \alpha) (\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}).$$

*Proof.* Choose  $\eta \in C_c^\infty(B_1)$  with  $\eta = 1$  on  $B_{3/4}$ , extend  $\eta f$  by zero, and set

$$w = \mathcal{N}(\eta f), \quad h = u + w, \quad -\Delta w = \eta f \implies \Delta h = 0 \text{ in } B_{3/4}.$$

The Newtonian formula and singular-integral estimate give

$$4.1.1 \wedge 4.1.2 \implies \|w\|_{C^{2,\alpha}(B_{3/4})} \leq C \|f\|_{C^{0,\alpha}(B_1)}.$$

The harmonic estimates of Theorems 2.1.17 and 2.1.20 yield

$$\|h\|_{C^{2,\alpha}(B_{1/2})} \leq C \|h\|_{L^\infty(B_{3/4})} \leq C (\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}), \quad u = h - w.$$

Combining the last two estimates proves 4.1.3.  $\square$

**Theorem 4.1.11** (Boundary Schauder estimate). *Write  $B_1^+ = B_1 \cap \{x_n > 0\}$  and  $B_1' = B_1 \cap \{x_n = 0\}$ . If  $u \in C^2(B_1^+) \cap C(\overline{B_1^+})$  and*

$$\begin{cases} \Delta u = f & \text{in } B_1^+, \\ u = \phi & \text{on } B_1', \end{cases} \quad f \in C^{0,\alpha}(\overline{B_1^+}), \quad \phi \in C^{2,\alpha}(B_1'),$$

*then*

$$\|u\|_{C^{2,\alpha}(\overline{B_{1/2}^+})} \leq C(n, \alpha) (\|u\|_{L^\infty(B_1^+)} + \|f\|_{C^{0,\alpha}(\overline{B_1^+})} + \|\phi\|_{C^{2,\alpha}(B_1')}).$$

**Proposition 4.1.12** (Campanato characterization). *Let  $0 < \alpha < 1$  and  $k \in \mathbb{Z}_{\geq 0}$ . With  $\mathcal{P}_k$  the polynomials of degree at most  $k$ , one has*

$$u \in C_{\text{loc}}^{k,\alpha}(\Omega) \iff \forall \Omega' \Subset \Omega, \quad \sup_{\substack{x \in \Omega' \\ 0 < r < \frac{1}{2} \text{dist}(\Omega', \partial\Omega)}} r^{-k-\alpha} \inf_{P \in \mathcal{P}_k} \|u - P\|_{L^\infty(B_r(x))} < \infty.$$

**Theorem 4.1.13** (Campanato iteration). *Let  $\Delta u = f$  in  $B_1$ ,  $f \in C^{0,\alpha}(B_1)$ , and*

$$\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)} \leq \delta(n, \alpha).$$

*There exist  $\rho = \rho(n, \alpha) \in (0, 1/2)$  and quadratic polynomials  $P_k$  such that*

$$\Delta P_k = f(0), \quad \|u - P_k\|_{L^\infty(B_{\rho^k})} \leq \rho^{k(2+\alpha)} \quad \forall k \geq 0.$$

*Together with Proposition 4.1.12, this recovers the local estimate 4.1.3.*

**Definition 4.1.14** (Kelvin transform). *For  $I(x) = x/|x|^2$  and  $\Omega^* = \{x \neq 0 : I(x) \in \Omega\}$ , define*

$$u^*(x) = |x|^{2-n} u(I(x)).$$

**Proposition 4.1.15** (Kelvin covariance). *If  $\Delta u = f$  in  $\Omega$ , then*

$$\Delta u^*(x) = |x|^{-n-2} f(I(x)) \quad \forall x \in \Omega^*.$$

*In particular,*

$$\Delta u = 0 \text{ in } \Omega \implies \Delta u^* = 0 \text{ in } \Omega^*.$$

## 5. SCHAUDER ESTIMATES

The Laplacian estimates of Section 4 extend to uniformly elliptic operators by freezing coefficients, rescaling, and flattening the boundary.

## 5.1. Variable Coefficients.

**Lemma 5.1.1** (Small perturbations). *Let  $0 < \alpha < 1$ , let  $u \in C^{2,\alpha}(B_1)$ , and let*

$$Lu = a^{ij}u_{ij} + b^i u_i + cu = f$$

*satisfy 1.1.2. There exists  $\delta_0 = \delta_0(n, \alpha, \lambda, \Lambda) > 0$  such that*

$$\|a^{ij} - \delta^{ij}\|_{C^{0,\alpha}(B_1)} + \|b\|_{C^{0,\alpha}(B_1)} + \|c\|_{C^{0,\alpha}(B_1)} \leq \delta_0 \implies \|u\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}).$$

*This follows from 4.1.3, Hölder interpolation, and a concentric-ball absorption.*

**Theorem 5.1.2** (Interior Schauder estimate). *Let  $L$  satisfy 1.1.2 in  $B_1$ , let  $a^{ij}, b^i, c \in C^{0,\alpha}(B_1)$ , and let  $u \in C^{2,\alpha}(B_1)$  satisfy  $Lu = f \in C^{0,\alpha}(B_1)$ . Then*

$$(5.1.1) \quad \|u\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}),$$

*where  $C$  depends only on  $n, \alpha, \lambda, \Lambda$  and the coefficient norms.*

*Proof.* Fix  $x_0 \in B_{1/2}$  and normalize  $A(x_0)$  to the identity by an affine change of variables. For  $v(y) = u(x_0 + ry)$ , the rescaled operator is

$$L_r v = a_r^{ij} v_{ij} + r b_r^i v_i + r^2 c_r v = r^2 f_r, \quad \Theta_r := \|a_r - I\|_{C^{0,\alpha}(B_1)} + r \|b_r\|_{C^{0,\alpha}(B_1)} + r^2 \|c_r\|_{C^{0,\alpha}(B_1)}.$$

$$\Theta_r \leq Cr^\alpha \|a\|_{C^{0,\alpha}} + Cr \|b\|_{C^{0,\alpha}} + Cr^2 \|c\|_{C^{0,\alpha}} \xrightarrow{r \downarrow 0} 0.$$

Choose  $r$  so that  $\Theta_r \leq \delta_0$ . Then

$$L_r v = r^2 f_r \xrightarrow{5.1.1} \|v\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|v\|_{L^\infty(B_1)} + r^2 \|f_r\|_{C^{0,\alpha}(B_1)}).$$

Scaling back gives

$$\|u\|_{C^{2,\alpha}(B_{r/2}(x_0))} \leq C_r(\|u\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}).$$

A finite cover of  $B_{1/2}$  proves 5.1.1. □

**Lemma 5.1.3** (Boundary flattening). *Suppose  $L$  satisfies 1.1.2 with  $a^{ij}, b^i, c \in C^{0,\alpha}$  and, locally,*

$$\Omega = \{(x', x_n) : x_n > g(x')\}, \quad g \in C^{2,\alpha}, \quad g(0) = 0, \quad \nabla g(0) = 0.$$

*The map  $\Phi(x) = (x', x_n - g(x'))$  sends the boundary to  $\{y_n = 0\}$ . If  $v = u \circ \Phi^{-1}$ , then*

$$Lu = f \iff \tilde{L}v = \tilde{a}^{k\ell} v_{k\ell} + \tilde{b}^k v_k + \tilde{c}v = \tilde{f}.$$

$$\tilde{a}^{k\ell}(\Phi(x)) = a^{ij}(x) \partial_i \Phi_k(x) \partial_j \Phi_\ell(x), \quad \tilde{b}^k(\Phi(x)) = a^{ij}(x) \partial_{ij} \Phi_k(x) + b^i(x) \partial_i \Phi_k(x).$$

*The transformed operator remains uniformly elliptic with  $C^{0,\alpha}$  coefficients.*

**Theorem 5.1.4** (Boundary Schauder estimate). *Set  $\Omega_r = \Omega \cap B_r$  and  $\Sigma_r = \partial\Omega \cap B_r$ , where  $\partial\Omega$  is  $C^{2,\alpha}$  near the origin. If  $L$  is uniformly elliptic with  $C^{0,\alpha}$  coefficients and*

$$\begin{cases} Lu = f & \text{in } \Omega_1, \\ u = \varphi & \text{on } \Sigma_1, \end{cases} \quad f \in C^{0,\alpha}(\overline{\Omega_1}), \quad \varphi \in C^{2,\alpha}(\Sigma_1), \quad u \in C^{2,\alpha}(\overline{\Omega_1}),$$

*then*

$$\|u\|_{C^{2,\alpha}(\overline{\Omega_{1/2}})} \leq C(\|u\|_{L^\infty(\Omega_1)} + \|f\|_{C^{0,\alpha}(\overline{\Omega_1})} + \|\varphi\|_{C^{2,\alpha}(\Sigma_1)}).$$

*The estimate follows by Lemma 5.1.3 and the flat half-ball perturbation estimate.*

**Theorem 5.1.5** (Global Schauder estimate). *Let  $\Omega$  be bounded with  $C^{2,\alpha}$  boundary, let  $L$  be uniformly elliptic with coefficients in  $C^{0,\alpha}(\overline{\Omega})$ , and let*

$$u \in C^{2,\alpha}(\overline{\Omega}), \quad f \in C^{0,\alpha}(\overline{\Omega}), \quad \varphi \in C^{2,\alpha}(\partial\Omega).$$

*Then*

$$(5.1.2) \quad \begin{cases} Lu = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega \end{cases} \implies \|u\|_{C^{2,\alpha}(\overline{\Omega})} \leq C(\|u\|_{L^\infty(\Omega)} + \|f\|_{C^{0,\alpha}(\overline{\Omega})} + \|\varphi\|_{C^{2,\alpha}(\partial\Omega)}).$$

*If  $c \leq 0$ , then*

$$5.1.2 \wedge 1.1.10 \implies \|u\|_{C^{2,\alpha}(\overline{\Omega})} \leq C(\|f\|_{C^{0,\alpha}(\overline{\Omega})} + \|\varphi\|_{C^{2,\alpha}(\partial\Omega)}).$$

**Theorem 5.1.6** (Laplacian solvability). *Let  $\Omega$  be bounded with  $C^{2,\alpha}$  boundary. Then*

$$f \in C^{0,\alpha}(\overline{\Omega}), \quad \varphi \in C^{2,\alpha}(\partial\Omega) \implies \exists! u \in C^{2,\alpha}(\overline{\Omega}) : \begin{cases} \Delta u = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

**Theorem 5.1.7** (Continuity method). *Let  $X, Y$  be Banach spaces and let  $t \mapsto L_t \in \mathcal{B}(X, Y)$  be continuous. Suppose  $L_0$  is an isomorphism and*

$$(5.1.3) \quad \|u\|_X \leq C\|L_t u\|_Y \quad \forall (t, u) \in [0, 1] \times X,$$

*with  $C$  independent of  $t$ . Then every  $L_t$  is an isomorphism.*

*Proof.* Let

$$I = \{t \in [0, 1] : L_t \text{ is surjective}\}.$$

Estimate 5.1.3 gives injectivity, while

$$s \in I, \quad C\|L_t - L_s\|_{\mathcal{B}(X, Y)} < 1 \implies L_t = L_s(I + L_s^{-1}(L_t - L_s)) \implies t \in I$$

by the Neumann series; hence  $I$  is open. If  $t_j \in I$ ,  $t_j \rightarrow t$ , and  $L_{t_j} u_j = y$ , then

$$L_{t_j}(u_j - u_k) = (L_{t_k} - L_{t_j})u_k \xrightarrow{5.1.3} \|u_j - u_k\|_X \leq C\|(L_{t_k} - L_{t_j})u_k\|_Y \leq C^2\|L_{t_k} - L_{t_j}\|_{\mathcal{B}(X, Y)}\|y\|_Y \xrightarrow{j, k \rightarrow \infty} 0.$$

Thus  $u_j \rightarrow u$  in  $X$ , continuity gives  $L_t u = y$ , and  $I$  is closed. Therefore  $I = [0, 1]$ .  $\square$

**Theorem 5.1.8** (Dirichlet existence). *Let  $\Omega$  be bounded with  $C^{2,\alpha}$  boundary, let  $L$  be uniformly elliptic with coefficients in  $C^{0,\alpha}(\overline{\Omega})$ , and suppose  $c \leq 0$ . For every*

$$f \in C^{0,\alpha}(\overline{\Omega}), \quad \varphi \in C^{2,\alpha}(\partial\Omega),$$

*there exists a unique  $u \in C^{2,\alpha}(\overline{\Omega})$  satisfying*

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

*Proof.* Extend and subtract  $\varphi$ , and set

$$X = \{u \in C^{2,\alpha}(\overline{\Omega}) : u|_{\partial\Omega} = 0\}, \quad Y = C^{0,\alpha}(\overline{\Omega}), \quad L_t = (1-t)\Delta + tL.$$

The Laplacian theory and the uniform global estimate give

$$5.1.6 \implies L_0 : X \longrightarrow Y \text{ is an isomorphism.}$$

Moreover,

$$5.1.2 \wedge 1.1.10 \implies \|u\|_X \leq C\|L_t u\|_Y \xrightarrow{5.1.7} L_1 = L \text{ is an isomorphism.}$$

Restoring  $\varphi$  gives existence; Theorem 1.1.10 gives uniqueness.  $\square$

**Theorem 5.1.9** (Fredholm alternative). *Let  $\Omega$  and  $L$  satisfy the hypotheses of Theorem 5.1.8, without the condition  $c \leq 0$ . Exactly one of the following holds:*

- (1) Every  $f \in C^{0,\alpha}(\overline{\Omega})$  and  $\varphi \in C^{2,\alpha}(\partial\Omega)$  determine a unique solution  $u \in C^{2,\alpha}(\overline{\Omega})$ .  
 (2) There exists  $0 \neq u \in C^{2,\alpha}(\overline{\Omega})$  satisfying

$$\begin{cases} Lu = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

**Theorem 5.1.10** (Higher regularity). *Let  $k \in \mathbb{Z}_{\geq 0}$ , let  $L$  satisfy 1.1.2, let  $a^{ij}, b^i, c, f \in C^{k,\alpha}(\Omega)$ , and let  $u \in C_{\text{loc}}^{2,\alpha}(\Omega)$  satisfy  $Lu = f$ . Then*

$$u \in C_{\text{loc}}^{k+2,\alpha}(\Omega), \quad \|u\|_{C^{k+2,\alpha}(\Omega')} \leq C(\|u\|_{L^\infty(\Omega)} + \|f\|_{C^{k,\alpha}(\Omega)}) \quad \forall \Omega' \Subset \Omega.$$

In particular,

$$a^{ij}, b^i, c, f \in C^\infty(\Omega) \implies u \in C^\infty(\Omega).$$

**Theorem 5.1.11** (Continuous boundary data). *Retain the hypotheses on  $\Omega, L, f$  and the condition  $c \leq 0$  from Theorem 5.1.8, but assume only  $\varphi \in C(\partial\Omega)$ . There exists a unique solution satisfying*

$$u \in C_{\text{loc}}^{2,\alpha}(\Omega) \cap C(\overline{\Omega}), \quad \begin{cases} Lu = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

Indeed, if  $\varphi_j \rightarrow \varphi$  uniformly and  $u_j$  is the solution with boundary value  $\varphi_j$ , then

$$\begin{cases} L(u_j - u_k) = 0 & \text{in } \Omega, \\ u_j - u_k = \varphi_j - \varphi_k & \text{on } \partial\Omega \end{cases} \xrightarrow{1.1.10} \|u_j - u_k\|_{L^\infty(\Omega)} \leq \|\varphi_j - \varphi_k\|_{L^\infty(\partial\Omega)} \longrightarrow 0,$$

and 5.1.1 gives uniform local  $C^{2,\alpha}$  bounds and compactness in  $C^{2,\beta}$  for every  $0 < \beta < \alpha$ .

## 6. SINGULAR INTEGRALS AND $W^{2,p}$ ESTIMATES

The Calderón–Zygmund theorem converts  $L^p$  control of the equation into two weak derivatives of the solution.

### 6.1. Sobolev Estimates.

**Theorem 6.1.1** ( $W^{2,p}$  Laplacian estimate). *Let  $1 < p < \infty$ , let  $u \in W^{2,p}(B_1)$ , and suppose  $\Delta u = f \in L^p(B_1)$ . Then*

$$(6.1.1) \quad \|u\|_{W^{2,p}(B_{1/2})} \leq C(n, p) (\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}).$$

*Proof.* Extend  $f$  by zero, set  $w = \mathcal{N}f = \Gamma * f$ , and let  $h = u + w$ . Since  $-\Delta w = f$ ,

$$\Delta h = \Delta u + \Delta w = f - f = 0, \quad u = h - w.$$

Moreover,

$$\Gamma, \nabla \Gamma \in L^1(B_2), \quad 4.1.1 \wedge 4.1.9 \implies \|w\|_{W^{2,p}(B_1)} \leq C(n, p) \|f\|_{L^p(B_1)}.$$

The harmonic estimates now give

$$\Delta h = 0 \xrightarrow{2.1.4, 2.1.17} \|h\|_{W^{2,p}(B_{1/2})} \leq C\|h\|_{L^p(B_1)} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}), \quad u = h - w,$$

which proves 6.1.1.  $\square$

**Lemma 6.1.2** (Small perturbations). *Let  $1 < p < \infty$ ,  $u \in W^{2,p}(B_1)$ , and  $Lu = f \in L^p(B_1)$ . There exists  $\delta_0 = \delta_0(n, p) > 0$  such that*

$$\|a^{ij} - \delta^{ij}\|_{L^\infty(B_1)} + \|b\|_{L^\infty(B_1)} + \|c\|_{L^\infty(B_1)} \leq \delta_0 \implies \|u\|_{W^{2,p}(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}).$$

This is the absorption of

$$Lu = f \xLeftrightarrow{1.1.9} \Delta u = f + (\delta^{ij} - a^{ij})u_{ij} - b^i u_i - cu$$

in 6.1.1, using Proposition 6.1.6.

**Theorem 6.1.3** (Interior estimate). *Let  $1 < p < \infty$ , let  $L$  satisfy 1.1.2 in  $B_1$ , let  $a^{ij}$  be continuous, and let  $b^i, c \in L^\infty(B_1)$ . If  $u \in W^{2,p}(B_1)$  and  $Lu = f \in L^p(B_1)$ , then*

$$\|u\|_{W^{2,p}(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}).$$

*The constant depends on  $n, p, \lambda, \Lambda$ , the lower-order bounds, and the modulus of continuity of  $A$ . The proof freezes  $A(x_0)$ , rescales, applies Lemma 6.1.2, and covers  $B_{1/2}$ .*

**Theorem 6.1.4** (Boundary estimate). *Write*

$$B_1^+ = B_1 \cap \{x_n > 0\}, \quad B_1' = B_1 \cap \{x_n = 0\}.$$

*Let  $1 < p < \infty$ , let  $L$  satisfy 1.1.2 with  $A \in C(\overline{B_1^+})$  and  $b^i, c \in L^\infty(B_1^+)$ . If*

$$\begin{cases} Lu = f & \text{in } B_1^+, \\ u = 0 & \text{on } B_1', \end{cases} \quad u \in W^{2,p}(B_1^+), \quad f \in L^p(B_1^+),$$

*then*

$$\|u\|_{W^{2,p}(B_{1/2}^+)} \leq C(\|u\|_{L^p(B_1^+)} + \|f\|_{L^p(B_1^+)}).$$

**Theorem 6.1.5** (Global estimate). *Let  $1 < p < \infty$ , let  $\Omega$  be a bounded  $C^{1,1}$  domain, and let  $L$  satisfy 1.1.2 with  $A \in C(\overline{\Omega})$  and  $b^i, c \in L^\infty(\Omega)$ . If  $u, \phi \in W^{2,p}(\Omega)$ ,  $u - \phi \in W_0^{1,p}(\Omega)$ , and*

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = \phi & \text{on } \partial\Omega, \end{cases} \quad f \in L^p(\Omega),$$

*then*

$$(6.1.2) \quad \|u\|_{W^{2,p}(\Omega)} \leq C(\|u\|_{L^p(\Omega)} + \|f\|_{L^p(\Omega)} + \|\phi\|_{W^{2,p}(\Omega)}).$$

*Proof.* Replace  $u$  by  $u - \phi$  and  $f$  by  $f - L\phi$ . Interior charts use Theorem 6.1.3;  $C^{1,1}$  boundary charts use the flattening computation of Lemma 5.1.3 and Theorem 6.1.4. Thus a finite-overlap cover  $\{U_j\}$  gives

$$\sum_j \|u\|_{W^{2,p}(U_j \cap \Omega)}^p \leq C \sum_j (\|u\|_{L^p(2U_j \cap \Omega)}^p + \|f\|_{L^p(2U_j \cap \Omega)}^p) \leq C(\|u\|_{L^p(\Omega)}^p + \|f\|_{L^p(\Omega)}^p).$$

Restoring  $\phi$  proves 6.1.2. □

**Proposition 6.1.6** (Interpolation). *Let  $1 < p < \infty$  and let  $\Omega$  be bounded and Lipschitz. For every  $\varepsilon > 0$ ,*

$$\|\nabla u\|_{L^p(\Omega)} \leq \varepsilon \|D^2 u\|_{L^p(\Omega)} + C(\varepsilon, p, \Omega) \|u\|_{L^p(\Omega)} \quad \forall u \in W^{2,p}(\Omega).$$

**Definition 6.1.7** (Contact set). *For  $u \in C(\overline{\Omega})$ , its upper contact set is*

$$\Gamma^+(u) = \{x \in \Omega : u(y) \leq u(x) + \mathbf{p} \cdot (y - x) \text{ for some } \mathbf{p} \in \mathbb{R}^n \text{ and every } y \in \overline{\Omega}\}.$$

*At every differentiability point in  $\Gamma^+(u)$ , one has  $\mathbf{p} = \nabla u(x)$ .*

**Theorem 6.1.8** (Alexandrov–Bakelman–Pucci). *Let  $\Omega$  be bounded, let  $d = \text{diam}(\Omega)$ , and let*

$$u \in W_{\text{loc}}^{2,n}(\Omega) \cap C(\overline{\Omega}).$$

*Suppose  $L$  satisfies 1.1.2,  $b \in L^n(\Omega)$ ,  $c \leq 0$ , and  $Lu \geq f$  almost everywhere. Then*

$$(6.1.3) \quad \sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + \frac{Cd}{\lambda} \|f^-\|_{L^n(\Gamma^+(u))}, \quad C = C(n, \lambda, \Lambda, d, \|b\|_{L^n(\Omega)}).$$

**Corollary 6.1.9** ( $W^{2,p}$  uniqueness). *Let  $\Omega$  be bounded and  $C^{1,1}$ , let  $L$  satisfy 1.1.2 with  $b^i, c \in L^\infty(\Omega)$  and  $c \leq 0$ , and let  $p > n$ . For  $u \in W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega)$ ,*

$$\begin{cases} Lu = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \xrightarrow{6.1.8} \sup_{\Omega} u \leq 0 \quad \wedge \quad \sup_{\Omega} (-u) \leq 0 \implies u \equiv 0.$$



**Theorem 6.1.10** ( $W^{2,p}$  solvability). *Let  $p > n$ , let  $\Omega$  be bounded with  $C^{2,\alpha}$  boundary, let  $L$  satisfy 1.1.2, let  $a^{ij} \in C(\overline{\Omega})$ , let  $b^i, c \in L^\infty(\Omega)$ , and suppose  $c \leq 0$ . For every*

$$f \in L^p(\Omega), \quad \varphi \in W^{2,p}(\Omega),$$

*there exists a unique  $u \in W^{2,p}(\Omega)$  satisfying*

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega, \end{cases} \quad u - \varphi \in W_0^{1,p}(\Omega),$$

*and*

$$\|u\|_{W^{2,p}(\Omega)} \leq C(\|f\|_{L^p(\Omega)} + \|\varphi\|_{W^{2,p}(\Omega)}).$$

*Proof.* Subtract  $\varphi$  and set

$$X_p = W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega), \quad Y_p = L^p(\Omega), \quad L_t = (1-t)\Delta + tL.$$

For  $L_t u = g$ , the ABP and global estimates give, uniformly in  $t$ ,

$$\begin{cases} L_t u = g & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \xrightarrow{6.1.3} \|u\|_{L^\infty(\Omega)} \leq C\|g\|_{L^n(\Omega)} \leq C\|g\|_{L^p(\Omega)}.$$

Consequently,

$$L_t u = g \xrightarrow{6.1.2} \|u\|_{W^{2,p}(\Omega)} \leq C(\|u\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)}) \leq C\|g\|_{L^p(\Omega)}.$$

For smooth  $g_j \rightarrow g$  in  $L^p$ , Theorem 5.1.8 gives solutions of  $\Delta u_j = g_j$  with zero boundary data, and

$$\|u_j - u_k\|_{W^{2,p}(\Omega)} \leq C\|g_j - g_k\|_{L^p(\Omega)} \implies L_0 : X_p \longrightarrow Y_p \text{ is an isomorphism.}$$

Therefore 5.1.3 holds on  $X_p \rightarrow Y_p$ , and

$$L_0 \text{ is an isomorphism } \xrightarrow{5.1.7} L_1 = L \text{ is an isomorphism.}$$

Restoring  $\varphi$  proves existence and the estimate; Corollary 6.1.9 gives uniqueness.  $\square$

**Corollary 6.1.11** (Higher regularity). *Let  $1 < p < \infty$ , let  $k \in \mathbb{Z}_{\geq 1}$ , let  $\Omega$  be bounded with  $\partial\Omega \in C^{k+1,1}$ , and let  $L$  satisfy 1.1.2. Assume*

$$a^{ij}, b^i, c \in W^{k,\infty}(\Omega), \quad f \in W^{k,p}(\Omega), \quad \varphi \in W^{k+2,p}(\Omega).$$

*If*

$$u \in W^{2,p}(\Omega), \quad \begin{cases} Lu = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega, \end{cases}$$

*then*

$$u \in W^{k+2,p}(\Omega), \quad \|u\|_{W^{k+2,p}(\Omega)} \leq C(\|u\|_{L^p(\Omega)} + \|f\|_{W^{k,p}(\Omega)} + \|\varphi\|_{W^{k+2,p}(\Omega)}).$$

**Theorem 6.1.12** (Fredholm alternative). *Retain the hypotheses of Theorem 6.1.10 but allow arbitrary  $c \in L^\infty(\Omega)$ . For*

$$X_p = W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega), \quad L : X_p \longrightarrow L^p(\Omega),$$

*the operator  $L$  is Fredholm of index zero. Consequently,*

$$\ker L = \{0\} \iff \operatorname{ran} L = L^p(\Omega) \iff \forall f \in L^p(\Omega) \exists! u \in X_p : Lu = f,$$

*whereas failure of these conditions gives*

$$1 \leq \dim \ker L = \operatorname{codim} \operatorname{ran} L < \infty.$$

## 7. SOBOLEV SPACES

Definition 1.1.5 and Theorems 1.1.6–1.1.8 contain the basic theory. We record only the calculus and boundary tools needed below.

### 7.1. Core Tools.

**Proposition 7.1.1** (Mollification). *Let  $u \in W^{k,p}(\Omega)$ ,  $1 \leq p < \infty$ , let  $\rho_\varepsilon$  be a standard mollifier, and let  $\Omega' \Subset \Omega$ . For sufficiently small  $\varepsilon$ ,*

$$(7.1.1) \quad D^\alpha(u * \rho_\varepsilon) = (D^\alpha u) * \rho_\varepsilon, \quad \|u * \rho_\varepsilon - u\|_{W^{k,p}(\Omega')} \longrightarrow 0 \quad (|\alpha| \leq k).$$

*Proof.* For  $\varphi \in C_c^\infty(\Omega')$ , Fubini and 1.1.1 give

$$\int_{\Omega'} (u * \rho_\varepsilon) D^\alpha \varphi = \int_{\Omega} u(y) (\rho_\varepsilon * D^\alpha \varphi)(y) dy = (-1)^{|\alpha|} \int_{\Omega'} ((D^\alpha u) * \rho_\varepsilon) \varphi,$$

while translation continuity gives

$$\|(D^\alpha u) * \rho_\varepsilon - D^\alpha u\|_{L^p(\Omega')} \leq \sup_{|h| \leq \varepsilon} \|D^\alpha u(\cdot - h) - D^\alpha u\|_{L^p(\Omega')} \longrightarrow 0.$$

Summing over  $|\alpha| \leq k$  proves 7.1.1. □

**Proposition 7.1.2** (Weak stability). *For  $1 < p < \infty$ ,*

$$u_j \rightharpoonup u \text{ in } L^p(\Omega), \quad D^\alpha u_j \rightharpoonup v \text{ in } L^p(\Omega) \implies D^\alpha u = v.$$

**Proposition 7.1.3** (Sobolev chain rule). *Let  $u \in W^{1,p}(\Omega)$  and let  $\phi \in C^1(\mathbb{R})$  satisfy  $\phi(0) = 0$  and  $\|\phi'\|_{L^\infty} < \infty$ . Then*

$$\phi(u) \in W^{1,p}(\Omega), \quad D_i(\phi(u)) = \phi'(u) D_i u \quad \text{a.e. in } \Omega.$$

**Corollary 7.1.4.** *For  $u \in W^{1,p}(\Omega)$ ,*

$$u^+ = \max\{u, 0\} \in W^{1,p}(\Omega), \quad D_i u^+ = \chi_{\{u>0\}} D_i u \quad \text{a.e.}$$

**Theorem 7.1.5** (Smooth approximation). *For every open  $\Omega \subset \mathbb{R}^n$ ,  $1 \leq p < \infty$ , and  $k \in \mathbb{Z}_{\geq 0}$ ,*

$$\overline{C^\infty(\Omega) \cap W^{k,p}(\Omega)}^{W^{k,p}(\Omega)} = W^{k,p}(\Omega), \quad \overline{C_c^\infty(\Omega)}^{W^{k,p}(\Omega)} = W_0^{k,p}(\Omega).$$

*The first assertion follows from a locally finite partition of unity and 7.1.1; the second is the definition of  $W_0^{k,p}$ .*

**Theorem 7.1.6** (Extension). *If  $\Omega$  is bounded and Lipschitz, then for every  $k \in \mathbb{Z}_{\geq 0}$  and  $1 \leq p \leq \infty$  there is a bounded linear map*

$$E : W^{k,p}(\Omega) \longrightarrow W^{k,p}(\mathbb{R}^n), \quad Eu|_\Omega = u, \quad \|Eu\|_{W^{k,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

**Theorem 7.1.7** (Trace). *Let  $\Omega$  be bounded and Lipschitz and let  $1 \leq p < \infty$ . There is a unique bounded linear trace map satisfying*

$$(7.1.2) \quad T : W^{1,p}(\Omega) \longrightarrow L^p(\partial\Omega), \quad Tu = u|_{\partial\Omega} \quad (u \in C(\overline{\Omega}) \cap W^{1,p}(\Omega)), \quad \|Tu\|_{L^p(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

*Moreover,*

$$u \in W_0^{1,p}(\Omega) \xLeftrightarrow{7.1.7} Tu = 0.$$

*Proof.* Flattening and a partition of unity reduce the estimate to  $\mathbb{R}_+^n$ . Choose  $\eta \in C_c^1([0, \infty))$  with  $\eta(0) = 1$ . For  $u \in C_c^1(\mathbb{R}_+^n)$ ,

$$|u(x', 0)| \leq \int_0^\infty (|\eta'(t)| |u(x', t)| + \eta(t) |D_n u(x', t)|) dt.$$

Minkowski and Hölder yield

$$\|u(\cdot, 0)\|_{L^p(\mathbb{R}^{n-1})} \leq C (\|u\|_{L^p(\mathbb{R}_+^n)} + \|D_n u\|_{L^p(\mathbb{R}_+^n)}) \leq C \|u\|_{W^{1,p}(\mathbb{R}_+^n)} \xrightarrow{7.1.6} \|Tu\|_{L^p(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

Theorems 7.1.6 and 7.1.5 prove 7.1.2 by density. Continuity gives  $W_0^{1,p}(\Omega) \subset \ker T$ ; if  $Tu = 0$ , local zero extension, mollification, and a partition of unity give  $u \in W_0^{1,p}(\Omega)$ . □

**Definition 7.1.8** (Differences). For  $e_i$  the  $i$ th coordinate vector and  $h \neq 0$ , set

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}.$$

**Theorem 7.1.9** (Difference quotients). Let  $1 < p \leq \infty$  and  $\Omega' \Subset \Omega$ . For all sufficiently small  $|h|$ ,

$$(7.1.3) \quad u \in W^{1,p}(\Omega) \implies \|D_i^h u\|_{L^p(\Omega')} \leq \|D_i u\|_{L^p(\Omega)}, \quad \sup_{0 < |h| < h_0} \|D_i^h u\|_{L^p(\Omega')} \leq K \implies \|D_i u\|_{L^p(\Omega')} \leq K.$$

*Proof.* The fundamental theorem of calculus and Jensen give

$$D_i^h u(x) = \int_0^1 D_i u(x + t h e_i) dt, \quad \|D_i^h u\|_{L^p(\Omega')}^p \leq \int_0^1 \int_{\Omega'} |D_i u(x + t h e_i)|^p dx dt \leq \|D_i u\|_{L^p(\Omega)}^p.$$

Conversely, take  $h_j \rightarrow 0$  with  $D_i^{h_j} u \rightharpoonup v$  in  $L^p$ , weak-\* when  $p = \infty$ . For every  $\varphi \in C_c^\infty(\Omega')$ ,

$$\int_{\Omega'} v \varphi = \lim_{j \rightarrow \infty} \int_{\Omega'} D_i^{h_j} u \varphi = - \lim_{j \rightarrow \infty} \int_{\Omega'} u D_i^{-h_j} \varphi = - \int_{\Omega'} u D_i \varphi, \quad \|v\|_{L^p(\Omega')} \leq \liminf_{j \rightarrow \infty} \|D_i^{h_j} u\|_{L^p(\Omega')} \leq K.$$

Thus  $v = D_i u$ , proving 7.1.3.  $\square$

**Corollary 7.1.10** ( $W^{1,\infty} = C^{0,1}$ ). If  $\Omega$  is bounded, connected, and Lipschitz, then every  $W^{1,\infty}$  class has a Lipschitz representative and

$$W^{1,\infty}(\Omega) = C^{0,1}(\overline{\Omega}), \quad \|u\|_{C^{0,1}(\overline{\Omega})} \asymp \|u\|_{W^{1,\infty}(\Omega)}.$$

Indeed, 7.1.3 gives the Lipschitz bound after extension by Theorem 7.1.6; the converse follows by testing bounded difference quotients against  $C_c^\infty(\Omega)$ .

## 8. THE $H^1$ THEORY

The divergence-form theory requires only measurable coefficients: existence follows from coercivity, while energy estimates replace pointwise differentiation.

**8.1. Energy Methods.** Let  $\Omega \subset \mathbb{R}^n$  be bounded and Lipschitz, and let  $A = (a^{ij}) \in L^\infty(\Omega)$  satisfy

$$\lambda |\xi|^2 \leq a^{ij}(x) \xi_i \xi_j, \quad |A(x)\xi| \leq \Lambda |\xi| \quad \text{for a.e. } x \in \Omega, \quad \xi \in \mathbb{R}^n.$$

Set

$$\mathcal{L}u := -\partial_i(a^{ij}\partial_j u), \quad B[u, v] := \int_{\Omega} a^{ij}\partial_j u \partial_i v dx, \quad H^{-1}(\Omega) := (H_0^1(\Omega))^*.$$

**Theorem 8.1.1** (Lax–Milgram). Let  $H$  be a real Hilbert space and let  $B : H \times H \rightarrow \mathbb{R}$  be bilinear. If

$$|B(u, v)| \leq M \|u\|_H \|v\|_H, \quad B(u, u) \geq \lambda \|u\|_H^2,$$

then every  $F \in H^*$  determines a unique  $u \in H$  such that

$$B(u, v) = F(v) \quad \forall v \in H, \quad \|u\|_H \leq \lambda^{-1} \|F\|_{H^*}.$$

*Proof.* Riesz representation defines  $T \in \mathcal{B}(H)$  by  $B(u, v) = (Tu, v)_H$ . Coercivity gives

$$\lambda \|u\|_H^2 \leq (Tu, u)_H \leq \|Tu\|_H \|u\|_H \implies \|Tu\|_H \geq \lambda \|u\|_H \implies \ker T = 0, \quad \text{ran } T \text{ closed.}$$

If  $w \perp \text{ran } T$ , then

$$0 = (Tw, w)_H = B(w, w) \geq \lambda \|w\|_H^2 \implies w = 0 \implies \text{ran } T = H.$$

Writing  $F(v) = (g, v)_H$  and taking  $u = T^{-1}g$  gives the claim and its estimate.  $\square$

**Definition 8.1.2** (Weak solutions). For  $f \in H^{-1}(\Omega)$  and  $\varphi \in H^1(\Omega)$ , a function  $u \in H^1(\Omega)$  weakly solves the Dirichlet problem precisely when

$$(8.1.1) \quad \begin{cases} \mathcal{L}u = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega \end{cases} \iff u - \varphi \in H_0^1(\Omega), \quad B[u, v] = \langle f, v \rangle \quad \forall v \in H_0^1(\Omega).$$

**Theorem 8.1.3** (Weak solvability). For every  $f \in H^{-1}(\Omega)$  and  $\varphi \in H^1(\Omega)$ ,

$$(8.1.2) \quad \exists! u \in H^1(\Omega) : \begin{cases} \mathcal{L}u = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega, \end{cases} \quad \|u\|_{H^1(\Omega)} \leq C(\|f\|_{H^{-1}(\Omega)} + \|\varphi\|_{H^1(\Omega)}).$$

*Proof.* Set  $w = u - \varphi$  and

$$F(v) := \langle f, v \rangle - B[\varphi, v], \quad |F(v)| \leq (\|f\|_{H^{-1}} + \Lambda \|\nabla \varphi\|_{L^2}) \|v\|_{H_0^1}, \quad B[v, v] \geq \lambda \|v\|_{H_0^1}^2.$$

Thus

$$B[w, v] = F(v) \xrightarrow{8.1.1} \exists! w \in H_0^1(\Omega), \quad \|w\|_{H_0^1} \leq \lambda^{-1} (\|f\|_{H^{-1}} + \Lambda \|\nabla \varphi\|_{L^2}),$$

and Theorem 1.1.7 yields 8.1.2.  $\square$

**Theorem 8.1.4** (Dirichlet principle). Assume  $A = A^T$  and let  $u$  be the solution in 8.1.3. Then

$$\mathcal{E}(v) := \frac{1}{2} B[v, v] - \langle f, v \rangle, \quad u = \operatorname{argmin}_{v - \varphi \in H_0^1(\Omega)} \mathcal{E}(v), \quad \mathcal{E}(u + w) - \mathcal{E}(u) = \frac{1}{2} B[w, w] \geq \frac{\lambda}{2} \|w\|_{H_0^1}^2.$$

**Theorem 8.1.5** (Caccioppoli inequality). Suppose  $u \in H^1(B_R)$  weakly satisfies

$$-\partial_i(a^{ij}\partial_j u) = f + \partial_i F^i \quad \text{in } B_R, \quad f \in L^2(B_R), \quad F \in L^2(B_R; \mathbb{R}^n).$$

Then, for  $0 < r < R$  and  $k \in \mathbb{R}$ ,

$$(8.1.3) \quad \int_{B_r} |\nabla u|^2 dx \leq \frac{C}{(R-r)^2} \int_{B_R} |u - k|^2 dx + C(R-r)^2 \int_{B_R} |f|^2 dx + C \int_{B_R} |F|^2 dx.$$

*Proof.* Choose  $\eta \in C_0^\infty(B_R)$  with  $\eta = 1$  on  $B_r$  and  $|\nabla \eta| \leq 2(R-r)^{-1}$ , and test 8.1.1 with  $v = \eta^2(u - k)$ . Then

$$\lambda \int \eta^2 |\nabla u|^2 \leq 2\Lambda \int \eta |u - k| |\nabla u| |\nabla \eta| + \int |f| \eta^2 |u - k| + \int |F| (\eta^2 |\nabla u| + 2\eta |u - k| |\nabla \eta|).$$

Young's inequality gives

$$\lambda \int \eta^2 |\nabla u|^2 \leq \frac{\lambda}{2} \int \eta^2 |\nabla u|^2 + C \int |\nabla \eta|^2 |u - k|^2 + C(R-r)^2 \int |f|^2 + C \int |F|^2,$$

and absorption yields 8.1.3.  $\square$

**Proposition 8.1.6** (Weak maximum principle). If  $u \in H^1(\Omega)$  and

$$\begin{cases} \mathcal{L}u \leq 0 & \text{weakly in } \Omega, \\ u \leq 0 & \text{on } \partial\Omega, \end{cases} \xrightarrow{7.1.4, 7.1.7} u^+ \in H_0^1(\Omega),$$

then

$$0 \geq B[u, u^+] \geq \lambda \|u^+\|_{L^2}^2 \implies u^+ = 0 \implies u \leq 0 \quad \text{a.e. in } \Omega.$$

**Theorem 8.1.7** (Interior  $H^2$  regularity). If  $A \in C^{0,1}(B_1)$ ,  $f \in L^2(B_1)$ , and  $u \in H^1(B_1)$  weakly satisfies  $\mathcal{L}u = f$ , then

$$u \in H^2(B_{1/2}), \quad \|u\|_{H^2(B_{1/2})} \leq C(\|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)}).$$

Set  $M = \|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)}$ . Testing with  $-D_\ell^{-h}(\eta^2 D_\ell^h u)$  and using 7.1.3 gives

$$\|\eta \nabla D_\ell^h u\|_{L^2}^2 \leq C(\|\nabla u\|_{L^2}^2 + \|f\|_{L^2}^2) \xrightarrow{8.1.3} \sup_{0 < |h| \leq 1} \|\nabla D_\ell^h u\|_{L^2(B_{1/2})} \leq CM \xrightarrow{7.1.9} D^2 u \in L^2(B_{1/2}).$$

**Theorem 8.1.8** (Higher regularity). *Let  $k \geq 1$ . If  $A \in C^{k-1,1}(B_1)$ ,  $f \in H^{k-1}(B_1)$ , and  $\mathcal{L}u = f$  weakly, then*

$$u \in H^{k+1}(B_{1/2}), \quad \|u\|_{H^{k+1}(B_{1/2})} \leq C(\|u\|_{L^2(B_1)} + \|f\|_{H^{k-1}(B_1)}).$$

*In particular,*

$$A, f \in C^\infty(\Omega) \implies u \in H_{\text{loc}}^m(\Omega) \quad \forall m \implies u \in C^\infty(\Omega).$$

**Theorem 8.1.9** (Meyers estimate). *There exists  $\varepsilon = \varepsilon(n, \lambda, \Lambda) > 0$  such that, whenever  $2 \leq p \leq 2 + \varepsilon$ ,  $B_{2r} \Subset \Omega$ , and*

$$-\partial_i(a^{ij}\partial_j u) = \partial_i F^i \quad \text{weakly in } B_{2r}, \quad F \in L^p(B_{2r}; \mathbb{R}^n),$$

*one has*

$$\left( \int_{B_r} |\nabla u|^p dx \right)^{1/p} \leq C \left( \int_{B_{2r}} |\nabla u|^2 dx \right)^{1/2} + C \left( \int_{B_{2r}} |F|^p dx \right)^{1/p}.$$

*In particular,*

$$\mathcal{L}u = 0 \implies \nabla u \in L_{\text{loc}}^{2+\varepsilon}(\Omega).$$

## 9. DE GIORGI - NASH - MOSER THEOREM

Retain the measurable uniformly elliptic matrix  $A = (a^{ij})$  from Section 8 and write

$$\mathcal{L}u = -\partial_i(a^{ij}\partial_j u), \quad B[u, v] = \int_{\Omega} a^{ij}\partial_j u \partial_i v dx.$$

The homogeneous equation already contains the essential regularity mechanism.

**9.1. Hölder Theory.** A function  $u \in H_{\text{loc}}^1(\Omega)$  is a weak subsolution, respectively supersolution, when

$$B[u, \phi] \leq 0, \quad \text{respectively} \quad B[u, \phi] \geq 0, \quad 0 \leq \phi \in C_c^\infty(\Omega).$$

**Theorem 9.1.1** (Local boundedness). *Let  $\mathcal{L}u \leq 0$  weakly in  $B_{2R} \Subset \Omega$ . For every  $p > 0$  and  $0 < r < s \leq 2R$ ,*

$$\text{ess sup}_{B_r} u_+ \leq C_p \left( \frac{s}{s-r} \right)^{n/p} \left( \int_{B_s} (u_+)^p dx \right)^{1/p}, \quad C_p = C(n, \lambda, \Lambda, p).$$

*Proof.* Since  $u_+$  is again a weak subsolution, scaling reduces the case  $p = 2$  first to  $r = \frac{1}{2}$  and  $s = 1$ . Put

$$r_j = \frac{1}{2} + 2^{-j-1}, \quad k_j = M(1 - 2^{-j}), \quad w_j = (u - k_j)_+, \quad Y_j = \int_{B_{r_j}} w_j^2 dx.$$

Choose  $\eta_j = 1$  on  $B_{r_{j+1}}$ ,  $\eta_j = 0$  on  $\partial B_{r_j}$ , and  $|\nabla \eta_j| \leq C2^j$ . For some  $\chi > 1$ ,  $\delta = 1 - \chi^{-1} > 0$ , and  $b = b(n) > 1$ , testing with  $\eta_j^2 w_j$ , then applying Theorem 1.1.6 and Hölder, gives

$$(9.1.1) \quad \int |\nabla(\eta_j w_j)|^2 dx \leq C4^j Y_j, \quad |\{w_{j+1} > 0\}| \leq C4^j M^{-2} Y_j \implies Y_{j+1} \leq Cb^j M^{-2\delta} Y_j^{1+\delta}.$$

Taking  $M = CY_0^{1/2}$  in 9.1.1 gives  $Y_j \rightarrow 0$ , hence  $u_+ \leq M$  a.e. in  $B_{1/2}$ . Repeating with radii  $r + (s-r)2^{-j}$  proves the  $p = 2$  estimate. For  $p < 2$ , writing  $S(r) = \text{ess sup}_{B_r} u_+$  yields

$$\|u_+\|_{L^2(B_s)} \leq S(s)^{1-p/2} \|u_+\|_{L^p(B_s)}^{p/2} \implies S(r) \leq \frac{1}{2} S(s) + C(s-r)^{-n/p} \|u_+\|_{L^p(B_s)},$$

and iteration in  $r < s$  gives the claim; for  $p \geq 2$ , apply the energy–Sobolev iteration directly to  $(u_+)^{p/2}$ .  $\square$

**Lemma 9.1.2** (Measure-to-point). *Let  $v \in H^1(B_R)$  be a weak subsolution and let  $0 < \delta \leq 1$ . There is  $v = v(n, \lambda, \Lambda, \delta) \in (0, 1)$  such that*

$$\mathcal{L}v \leq 0 \text{ in } B_R, \quad v \leq 1, \quad |\{v \leq 0\} \cap B_R| \geq \delta |B_R| \implies \text{ess sup}_{B_{R/2}} v \leq 1 - v.$$

**Proposition 9.1.3** (Oscillation decay). *There are  $\theta \in (0, 1)$  and*

$$\alpha = \min \left\{ \frac{1}{2}, -\log_2 \theta \right\} \in (0, 1),$$

*depending only on  $n, \lambda, \Lambda$ , such that every weak solution of  $\mathcal{L}u = 0$  satisfies*

$$(9.1.2) \quad \operatorname{osc}_{B_{R/2}} u \leq \theta \operatorname{osc}_{B_R} u, \quad 0 < r \leq R \implies \operatorname{osc}_{B_r} u \leq C \left( \frac{r}{R} \right)^\alpha \operatorname{osc}_{B_R} u.$$

*Proof.* Set

$$m = \operatorname{ess\,inf}_{B_R} u, \quad M = \operatorname{ess\,sup}_{B_R} u, \quad \omega = M - m, \quad v = \frac{u - m}{\omega}$$

when  $\omega > 0$ . Then

$$|\{v \leq \frac{1}{2}\} \cap B_R| \geq \frac{1}{2}|B_R| \quad \vee \quad |\{v \geq \frac{1}{2}\} \cap B_R| \geq \frac{1}{2}|B_R|.$$

Apply Lemma 9.1.2 to  $2v - 1$  in the first case and to  $1 - 2v$  in the second:

$$\operatorname{ess\,sup}_{B_{R/2}} v \leq 1 - \frac{v}{2} \quad \vee \quad \operatorname{ess\,inf}_{B_{R/2}} v \geq \frac{v}{2} \implies \operatorname{osc}_{B_{R/2}} u \leq \left(1 - \frac{v}{2}\right) \omega.$$

Thus  $\theta = 1 - v/2$ ; dyadic iteration gives the second estimate in 9.1.2.  $\square$

**Theorem 9.1.4** (De Giorgi–Nash–Moser). *Every weak solution of  $\mathcal{L}u = 0$  belongs to  $C_{\text{loc}}^{0,\alpha}(\Omega)$ . More precisely,*

$$B_{2R} \subseteq \Omega \implies [u]_{C^{0,\alpha}(B_R)} \leq CR^{-\alpha} \left( \int_{B_{2R}} |u|^2 dx \right)^{1/2}, \quad \alpha = \alpha(n, \lambda, \Lambda) \in (0, 1).$$

*This follows from 9.1.2 and Theorem 9.1.1; the oscillation representatives are continuous.*

**Theorem 9.1.5** (Weak Harnack). *There are  $p_0 = p_0(n, \lambda, \Lambda) > 0$  and  $C = C(n, \lambda, \Lambda)$  such that*

$$(9.1.3) \quad u \geq 0, \quad \mathcal{L}u \geq 0 \text{ in } B_{2R} \implies \left( \int_{B_{3R/2}} u^{p_0} dx \right)^{1/p_0} \leq C \operatorname{ess\,inf}_{B_R} u.$$

**Theorem 9.1.6** (Harnack inequality). *If  $u \geq 0$  and  $\mathcal{L}u = 0$  weakly in  $B_{2R}$ , then*

$$\operatorname{ess\,sup}_{B_R} u \xrightarrow{9.1.1} C \left( \int_{B_{3R/2}} u^{p_0} dx \right)^{1/p_0} \xrightarrow{9.1.3} C \operatorname{ess\,inf}_{B_R} u.$$

*In particular, a nonnegative solution whose Hölder representative vanishes at one interior point vanishes identically on its connected component.*

## 10. MINIMAL SURFACE EQUATION

Minimal graphs are critical points of convex gradient functionals. Strict convexity gives uniqueness; gradient bounds, Section 9, and the Schauder theory give classical solvability.

**10.1. Convex Graphs.** Let  $F \in C^1(\mathbb{R}^n)$  be convex with  $|\nabla F(p)| \leq C(1 + |p|)$ . For boundary data  $\varphi \in T(H^1(B_1))$ , set

$$\mathcal{J}_F(v) := \int_{B_1} F(\nabla v) dx, \quad \mathcal{A}_\varphi := \{v \in H^1(B_1) : Tv = \varphi\}.$$

The class  $\mathcal{A}_\varphi$  is well-defined by Theorem 7.1.7.

**Proposition 10.1.1** (Euler–Lagrange equation). *For  $u \in \mathcal{A}_\varphi \cap C^1(\bar{B}_1)$ ,*

$$(10.1.1) \quad u \in \operatorname{argmin}_{v \in \mathcal{A}_\varphi} \mathcal{J}_F(v) \iff \int_{B_1} \nabla F(\nabla u) \cdot \nabla \eta dx = 0 \quad \forall \eta \in C_c^\infty(B_1).$$

*If  $F \in C^2(\mathbb{R}^n)$  and  $D^2F \geq \lambda I$ , the minimizer is unique.*

## 11. BOUNDARY PROBLEM

The small-slope model for a membrane constrained above an obstacle is

$$\sqrt{1+|p|^2} = 1 + \frac{1}{2}|p|^2 + \mathcal{O}(|p|^4), \quad \min_{v \geq \psi} \int_{\Omega} \left( \sqrt{1+|\nabla v|^2} + fv \right) dx \rightsquigarrow \min_{v \geq \psi} \int_{\Omega} \left( \frac{1}{2}|\nabla v|^2 + fv \right) dx.$$

The unknown coincidence set makes the Euler equation a free-boundary problem.

**11.1. Variational Problem.** Let  $\Omega \subset \mathbb{R}^n$  be bounded and Lipschitz, let  $f \in L^2(\Omega)$ , and let  $\psi, g \in H^1(\Omega)$  satisfy  $T\psi \leq Tg$ . Define

$$\mathcal{K}_{\psi,g} := \{v \in H^1(\Omega) : v - g \in H_0^1(\Omega), v \geq \psi \text{ a.e.}\}, \quad \mathcal{J}_f(v) := \int_{\Omega} \left( \frac{1}{2}|\nabla v|^2 + fv \right) dx.$$

By Corollary 7.1.4 and Theorem 7.1.7,  $g + (\psi - g)_+ \in \mathcal{K}_{\psi,g}$ , so the admissible class is nonempty.

**Theorem 11.1.1** (Variational inequality). *There is a unique minimizer  $u \in \mathcal{K}_{\psi,g}$ , characterized by*

$$(11.1.1) \quad u \in \operatorname{argmin}_{v \in \mathcal{K}_{\psi,g}} \mathcal{J}_f(v) \iff \int_{\Omega} (\nabla u \cdot \nabla(v - u) + f(v - u)) dx \geq 0 \quad (v \in \mathcal{K}_{\psi,g}).$$

*Proof.* Write  $v = g + w$ ,  $w \in H_0^1(\Omega)$ . Theorem 1.1.7 and Young's inequality give

$$\mathcal{J}_f(g + w) \geq \frac{1}{4} \|\nabla w\|_{L^2(\Omega)}^2 - C(\|g\|_{H^1(\Omega)}^2 + \|f\|_{L^2(\Omega)}^2).$$

Thus a minimizing sequence is bounded in  $H^1$ ; weak compactness, weak closedness of  $\mathcal{K}_{\psi,g}$ , and lower semicontinuity produce a minimizer. For  $v \in \mathcal{K}_{\psi,g}$ , the segment  $u + t(v - u)$  is admissible, whence

$$0 \leq \left. \frac{d}{dt} \right|_{t=0^+} \mathcal{J}_f(u + t(v - u)) = \int_{\Omega} (\nabla u \cdot \nabla(v - u) + f(v - u)) dx.$$

Conversely, 11.1.1 gives

$$\mathcal{J}_f(v) - \mathcal{J}_f(u) = \int_{\Omega} (\nabla u \cdot \nabla(v - u) + f(v - u)) dx + \frac{1}{2} \|\nabla(v - u)\|_{L^2(\Omega)}^2 \geq 0.$$

Two minimizers therefore have equal gradients; their common trace gives equality by Theorem 1.1.7.  $\square$

**Proposition 11.1.2** (Complementarity). *Assume  $u$  and  $\psi$  are continuous and set*

$$\Lambda(u) := \{u = \psi\}, \quad \Omega(u) := \{u > \psi\}, \quad \Gamma(u) := \partial\Omega(u) \cap \Omega.$$

*Then  $\mu := -\Delta u + f$  is a nonnegative Radon measure supported on  $\Lambda(u)$ ; equivalently,*

$$(11.1.2) \quad \begin{cases} u \geq \psi & \text{in } \Omega, \\ -\Delta u + f \geq 0 & \text{in } \Omega, \\ -\Delta u + f = 0 & \text{in } \Omega(u), \\ (u - \psi)(-\Delta u + f) = 0 & \text{in } \Omega, \\ Tu = Tg & \text{on } \partial\Omega. \end{cases}$$

*Indeed,*

$$0 \leq \eta \in C_c^\infty(\Omega) \xrightarrow{11.1.1} \langle \mu, \eta \rangle = \int_{\Omega} (\nabla u \cdot \nabla \eta + f\eta) dx \geq 0.$$

*Moreover,*

$$\operatorname{Supp}(\eta) \Subset \Omega(u) \implies \langle \mu, \pm \eta \rangle \geq 0 \implies \langle \mu, \eta \rangle = 0,$$

*which proves 11.1.2 and  $\operatorname{Supp}(\mu) \subset \Lambda(u)$ .*

**Theorem 11.1.3** (Optimal regularity). *Let  $0 < \alpha < 1$ ,  $f \in C_{\text{loc}}^{0,\alpha}(\Omega)$ , and  $\psi \in C_{\text{loc}}^{2,\alpha}(\Omega)$ . The minimizer in Theorem 11.1.1 satisfies*

$$u \in C_{\text{loc}}^{1,1}(\Omega), \quad \|u\|_{C^{1,1}(\Omega')} \leq C(\|u\|_{L^2(\Omega)} + \|f\|_{C^{0,\alpha}(\Omega')} + \|\psi\|_{C^{2,\alpha}(\Omega')})$$

whenever  $\Omega' \Subset \Omega'' \Subset \Omega$ . In particular, with

$$w := u - \psi, \quad h := f - \Delta\psi,$$

the complementarity system becomes

$$w \geq 0, \quad \Delta w = h \mathbf{1}_{\{w>0\}} \quad \text{a.e. in } \Omega, \quad \Gamma(u) = \partial\{w > 0\} \cap \Omega.$$

**Proposition 11.1.4** (Quadratic growth). *Assume the hypotheses of Theorem 11.1.3 and  $h \geq c_0 > 0$ . If  $x_0 \in \Gamma(u)$  and  $B_{2r}(x_0) \Subset \Omega$ , then*

$$(11.1.3) \quad \frac{c_0}{2n} r^2 \leq \sup_{B_r(x_0)} w \leq Cr^2, \quad \|\nabla w\|_{L^\infty(B_r(x_0))} \leq Cr.$$

*Proof.* Since  $w \geq 0$ ,  $w(x_0) = 0$ , and  $w \in C^{1,1}$ ,

$$\nabla w(x_0) = 0 \implies 0 \leq w(x) \leq \frac{1}{2} \|D^2 w\|_{L^\infty(B_{2r})} |x - x_0|^2, \quad |\nabla w(x)| \leq \|D^2 w\|_{L^\infty(B_{2r})} |x - x_0|.$$

Choose  $x_k \in \{w > 0\}$  with  $x_k \rightarrow x_0$  and put  $r_k = r - |x_k - x_0|$ . On each component of  $\{w > 0\} \cap B_{r_k}(x_k)$ ,

$$\Delta \left( w - \frac{c_0}{2n} |x - x_k|^2 \right) = h - c_0 \geq 0 \xrightarrow{8.1.6} \sup_{B_{r_k}(x_k)} w \geq w(x_k) + \frac{c_0}{2n} r_k^2.$$

Letting  $k \rightarrow \infty$  gives the lower bound in 11.1.3. □

**Theorem 11.1.5** (Free-boundary alternatives). *Assume  $h \in C^{0,\alpha}(\Omega)$  and  $h > 0$ . For  $x_0 \in \Gamma(u)$ , every sequence*

$$w_{x_0,r}(x) := \frac{w(x_0 + rx)}{r^2}, \quad r \downarrow 0,$$

has a locally  $C^1$  convergent subsequence whose limit is either

$$\frac{h(x_0)}{2} (x \cdot \nu)_+^2, \quad \nu \in \mathbb{S}^{n-1}, \quad \text{or} \quad \frac{1}{2} x \cdot Qx, \quad Q = Q^\top \geq 0, \quad \text{tr}(Q) = h(x_0).$$

The first alternative defines the relatively open regular set, where  $\Gamma(u)$  is locally  $C^{1,\beta}$  for some  $\beta \in (0, 1)$ ; all remaining points are singular.



## Part 2. Seminar talks<sup>4</sup>

### 12. SPECTRAL RIGIDITY OF RANDOM COVERS OF COMPACT HYPERBOLIC SURFACES

The universal cover of every closed hyperbolic surface is  $\mathbb{H}^2$ , whose spectrum is  $[\frac{1}{4}, \infty)$ . The large-cover problem asks whether this Plancherel spectrum also governs random finite covers.

**12.1. Random Covers.** Fix a compact connected orientable hyperbolic surface  $X = \Gamma \backslash \mathbb{H}^2$  of genus  $g \geq 2$ . Write

$$\Gamma = \langle a_1, \dots, a_{2g} \mid [a_1, a_2] \cdots [a_{2g-1}, a_{2g}] = 1 \rangle, \quad \mathcal{X}_{g,n} := \text{Hom}(\Gamma, S_n), \quad [n] := \{1, \dots, n\}.$$

**Definition 12.1.1** (Random cover). Choose  $\sigma_n \in \mathcal{X}_{g,n}$  uniformly and let  $\Gamma$  act by

$$\gamma \cdot (z, i) := (\gamma z, \sigma_n(\gamma)i), \quad X_n := \Gamma \backslash \sigma_n(\mathbb{H}^2 \times [n]), \quad \text{vol}(X_n) = 4\pi n(g-1).$$

The covering is connected precisely when  $\sigma_n(\Gamma)$  acts transitively:

$$X_n \text{ connected} \iff \sigma_n(\Gamma) \curvearrowright [n] \text{ transitive}, \quad X_n \text{ connected} \implies g(X_n) = n(g-1) + 1.$$

Let  $\rho_n$  be the permutation representation on  $V_n := \ell^2([n])$ . Its invariant constant line and zero-sum subspace give

$$\text{tr } \rho_n(\gamma) = \#\text{Fix}(\sigma_n(\gamma)), \quad V_n = \mathbb{C}\mathbf{1} \oplus V_n^0, \quad L^2(X_n) \cong L^2(X) \oplus L^2(X; V_n^0).$$

Here  $L^2(X; V_n^0)$  denotes sections of the associated flat bundle. Thus the constant line records the old spectrum and  $V_n^0$  records the new spectrum.

For the nonnegative Laplacian  $-\Delta_{X_n}$ , write

$$0 = \lambda_0(X_n) \leq \lambda_1(X_n) \leq \dots \longrightarrow \infty, \quad \lambda_j(X_n) = \frac{1}{4} + r_{j,n}^2, \quad r_{j,n} \in \mathbb{R} \cup i[0, \frac{1}{2}].$$

The Plancherel measure of  $\mathbb{H}^2$  is

$$d\mu_{\text{Pl}}(r) = \frac{1}{2\pi} r \tanh(\pi r) dr, \quad \lambda = \frac{1}{4} + r^2, \quad r \geq 0.$$

$$\mu_{\text{Pl}}([0, \sqrt{\Lambda - 1/4}]) = \frac{1}{2\pi} \int_0^{\sqrt{\Lambda - 1/4}} r \tanh(\pi r) dr.$$

For comparison, let  $Y_g$  be a closed genus- $g$  surface sampled with Weil–Petersson probability. For every  $\varepsilon > 0$ ,

$$\mathbb{P}_{\text{WP}}^g \left( \lambda_1(Y_g) > \frac{3}{16} - \varepsilon \right) \longrightarrow 1, \quad \mathbb{P}_{\text{WP}}^g \left( \lambda_1(Y_g) \geq \frac{1}{4} - \varepsilon \right) \longrightarrow 1.$$

The first limit is due independently to Wu–Xue and Lipnowski–Wright; Anantharaman–Monk proved the optimal second limit.

**Conjecture 12.1.2** (GOE universality). For  $\lambda_j(X_n) > \frac{1}{4}$ , unfold the spectrum by

$$\xi_{j,n} := n(2g-2) \int_0^{\sqrt{\lambda_j(X_n) - 1/4}} r \tanh(\pi r) dr.$$

Set  $s_{j,n} := \xi_{j+1,n} - \xi_{j,n}$ . The local empirical statistics of  $\{s_{j,n}\}$  are expected to converge to the GOE bulk-spacing law. Smooth number-variance limits are known in restricted iterated regimes, but full spacing convergence remains open.

Random regular graphs provide the discrete model. If  $G_N$  is a uniformly random simple  $d$ -regular graph, set

$$H_{G_N} := \frac{A_{G_N}}{\sqrt{d-1}}, \quad \theta_1 = \frac{d}{\sqrt{d-1}} \geq \theta_2 \geq \dots \geq \theta_N, \quad \hat{j} := \min\{j, N-j+1\}.$$

<sup>4</sup>Based on lectures by Elena Kim [1].

**Theorem 12.1.3** (Huang–McKenzie–Yau). *Let  $d \geq 3$  be fixed, let  $N \rightarrow \infty$  through values with  $dN$  even, and let  $\gamma_j^{(d)}$  be the Kesten–McKay classical locations. With probability  $1 - N^{-1+o(1)}$ ,*

$$|\theta_j - \gamma_j^{(d)}| \leq \frac{N^{o(1)}}{N^{2/3} \hat{1}^{1/3}}, \quad 2 \leq j \leq N.$$

Moreover, for  $a_d = d(d-1)/(d-2)^2$ ,

$$(a_d N)^{2/3}(\theta_2 - 2) \implies \text{TW}_1, \quad (a_d N)^{2/3}(-\theta_N - 2) \implies \text{TW}_1.$$

The classical locations are governed by the Kesten–McKay law, not the semicircle law:

local switchings  $\wedge$  resolvent equations  $\implies$  optimal local law  $\implies$  rigidity,  
microscopic loop equations  $\wedge$  Dyson Brownian motion  $\implies$  edge universality  $\implies$   $\text{TW}_1$ .

We now return to hyperbolic surfaces. For even  $\phi \in C_c^\infty(\mathbb{R})$ , set

$$\hat{\phi}(r) := \int_{\mathbb{R}} \phi(t) e^{-irt} dt, \quad k_\phi(t) := -\frac{1}{\sqrt{2\pi}} \int_t^\infty \frac{\phi'(s)}{\sqrt{\cosh s - \cosh t}} ds, \quad K_\phi(z, w) := k_\phi(d_{\mathbb{H}}(z, w)).$$

**Lemma 12.1.4** (Selberg formulas). *Let  $Y = \Gamma_Y \backslash \mathbb{H}^2$  be closed and have orthonormal eigenbasis  $\{u_j\}$ . The pre-trace formula is*

$$\sum_{j \geq 0} \hat{\phi}(r_j) |u_j(z)|^2 = \frac{1}{2\pi} \int_0^\infty \hat{\phi}(r) r \tanh(\pi r) dr + \sum_{\gamma \in \Gamma_Y \setminus \{1\}} K_\phi(z, \gamma z).$$

For the random cover in Definition 12.1.1, the trace formula becomes

$$(12.1.1) \quad \sum_{j \geq 0} \hat{\phi}(r_{j,n}) = \frac{n \text{vol}(X)}{2\pi} \int_0^\infty \hat{\phi}(r) r \tanh(\pi r) dr + \sum_{\gamma \in \mathcal{P}(X)} \sum_{m \geq 1} \frac{\ell_\gamma \phi(m \ell_\gamma)}{2 \sinh(m \ell_\gamma / 2)} \text{tr} \rho_n(\gamma^m),$$

where  $\mathcal{P}(X)$  denotes the primitive closed geodesics of  $X$ .

The geometric terms are controlled by short-loop geometry. If  $Y = \Gamma_Y \backslash \mathbb{H}^2$  and  $\text{inrad}_z(Y) \geq \delta$ , then

$$\#\{\gamma \in \Gamma_Y : d_{\mathbb{H}}(z, \gamma z) \leq L\} \leq C \delta^{-1} e^L, \quad 0 < \delta < 1,$$

while the Margulis lemma permits at most one primitive loop of length at most  $\varepsilon_0 := 2 \text{arcsinh}(1)$  based at  $z$ , up to inversion. Every cover also satisfies

$$\text{inrad}(X_n) \geq \text{inrad}(X),$$

so these estimates are uniform in  $n$ .

**Lemma 12.1.5** (Polynomial sampling). *Let  $P$  be a real polynomial with  $\deg P \leq q$ . For every  $k \in \mathbb{N}$ ,*

$$\sup_{0 \leq t \leq 1/(2q^2)} |P^{(k)}(t)| \leq \frac{2^{2k+1} q^{4k}}{(2k-1)!!} \sup_{m \geq q^2} \left| P\left(\frac{1}{m}\right) \right|, \quad (2k-1)!! := (2k-1)(2k-3) \cdots 1.$$

In particular,

$$P(0) = 0 \quad \wedge \quad n \geq 2q^2 \xrightarrow{12.1.5} \left| P\left(\frac{1}{n}\right) \right| \leq \frac{8q^4}{n} \sup_{m \geq q^2} \left| P\left(\frac{1}{m}\right) \right|.$$

**Proposition 12.1.6** (Polynomial moments). *For nontrivial words  $\gamma_1, \gamma_2 \in \Gamma$  of total length at most  $q$  and  $n \geq (Cq)^C$ , the random-cover moments admit*

$$\mathbb{E}[\text{tr} \rho_n(\gamma_1) \text{tr} \rho_n(\gamma_2)] = \frac{Q_{\gamma_1, \gamma_2}^{(2)}(1/n)}{R_{\gamma_1, \gamma_2}^{(2)}(1/n)} + O((Cq)^{Cq} n^{-q}), \quad R_{\gamma_1, \gamma_2}^{(2)}(1/n) \asymp 1,$$

$$\deg Q_{\gamma_1, \gamma_2}^{(2)} \leq 9q(4g+1), \quad \deg R_{\gamma_1, \gamma_2}^{(2)} \leq 9q(4g+1) + 1.$$

Define the off-diagonal quadruples by

$$\tilde{\Gamma}^4 := (\Gamma \setminus \{1\})^4 \setminus \left\{ (\gamma^{k_1}, \dots, \gamma^{k_4}) : \gamma \neq 1, k_\ell \in \mathbb{Z} \setminus \{0\} \right\}.$$

If  $\gamma = (\gamma_1, \dots, \gamma_8) \in (\tilde{\Gamma}^4)^2$  has total word length at most  $q$  and  $n \geq (Cq)^C$ , then

$$\mathbb{E} \left[ \sum_{i=1}^n \prod_{\ell=1}^8 \rho_n(\gamma_\ell)_{ii} \right] = \frac{Q_\gamma^{(8)}(1/n)}{R_\gamma^{(8)}(1/n)} + O((Cq)^{Cq} n^{-q-1}), \quad Q_\gamma^{(8)}(0) = 0, \quad R_\gamma^{(8)}(1/n) \asymp 1,$$

$$\max \left\{ \deg Q_\gamma^{(8)}, \deg R_\gamma^{(8)} \right\} \leq 9(q+1)(4g+1) + 1.$$

Consequently, Lemma 12.1.5 converts cancellation at  $n = \infty$  into polynomial decay in  $n$ .

Define

$$N_n(\Lambda) := \#\{j : \lambda_j(X_n) \leq \Lambda\}, \quad \int_0^{\sqrt{\lambda_{j,n}^{\mathbb{H}} - 1/4}} r \tanh(\pi r) dr = \frac{j}{n(2g-2)}.$$

**Theorem 12.1.7** (Kim–Tao, [1]). *For every  $\varepsilon > 0$ , there are  $\alpha = \alpha(g, \varepsilon) > 0$  and  $C = C(X) > 0$  such that, with probability at least  $1 - n^{-1/10}$ , every  $\Lambda \geq \frac{1}{4}$  and every  $j$  with  $\lambda_j(X_n) \in [\frac{1}{4}, \Lambda]$  satisfy*

$$(12.1.2) \quad \left| \lambda_j(X_n) - \lambda_{j,n}^{\mathbb{H}} \right| \leq C\Lambda^{1/2+\varepsilon} n^{-2\alpha/3}.$$

On the same event,

$$(12.1.3) \quad N_n(\Lambda) = (2g-2)n \int_0^{\sqrt{\Lambda-1/4}} r \tanh(\pi r) dr + O_{X,\varepsilon}(n^{1-\alpha} \Lambda^{1/2+\varepsilon}), \quad \Lambda \geq \frac{1}{4}.$$

**Corollary 12.1.8** (Plancherel convergence). *On the event in Theorem 12.1.7, for every  $R \geq 0$ ,*

$$\left| \frac{1}{n} N_n\left(\frac{1}{4} + R^2\right) - (2g-2) \int_0^R r \tanh(\pi r) dr \right| \leq C_{X,\varepsilon} n^{-\alpha} (1 + R^2)^{1/2+\varepsilon}.$$

Thus  $n^{-1}$  times the spectral counting measure converges locally, with polynomial error, to the pushforward of  $\text{vol}(X)\mu_{\text{Pl}}$  under  $r \mapsto \frac{1}{4} + r^2$ .

**Corollary 12.1.9** (Multiplicity). *On the same event,*

$$N_n\left(\frac{1}{4}\right) \leq Cn^{1-\alpha}, \quad \sup_{\lambda \in [1/4, \Lambda]} \dim \ker(-\Delta_{X_n} - \lambda) \leq C_{X,\varepsilon} n^{1-\alpha} \Lambda^{1/2+\varepsilon}.$$

*Proof.* The first estimate is 12.1.3 at  $\Lambda = \frac{1}{4}$ . For  $\lambda \in (\frac{1}{4}, \Lambda]$  and  $0 < \delta < \lambda - \frac{1}{4}$ ,

$$\dim \ker(-\Delta_{X_n} - \lambda) \leq N_n(\lambda + \delta) - N_n(\lambda - \delta) \stackrel{12.1.3}{\implies} Cn\delta + Cn^{1-\alpha} \Lambda^{1/2+\varepsilon}.$$

Letting  $\delta \downarrow 0$  proves the claim; the endpoint  $\lambda = \frac{1}{4}$  follows from the first estimate.  $\square$

**Theorem 12.1.10** (Eigenfunction delocalization, [1]). *For every  $\varepsilon > 0$ , there are  $\beta = \beta(g, \varepsilon) > 0$  and  $C = C(X) > 0$  such that, with probability at least  $1 - n^{-1/10}$ ,*

$$-\Delta_{X_n} u_j = \lambda_j(X_n) u_j, \quad \lambda_j(X_n) \leq \Lambda, \quad \Lambda \geq \frac{1}{4} \implies \|u_j\|_{L^\infty(X_n)} \leq C\Lambda^{1/4+\varepsilon} n^{-\beta} \|u_j\|_{L^2(X_n)}.$$

### Part 3. Appendix

#### APPENDIX A. FOUNDATIONAL TOOLS

##### A.1. Core Estimates.

**Proposition A.1.1** (Hölder, Minkowski, and Jensen). *Let  $1 \leq r, p, p_1, \dots, p_m \leq \infty$  and  $r^{-1} = \sum_{j=1}^m p_j^{-1}$ . Then*

$$\left\| \prod_{j=1}^m f_j \right\|_{L^r(X)} \leq \prod_{j=1}^m \|f_j\|_{L^{p_j}(X)}, \quad |\langle f, g \rangle| \leq \|f\|_{L^2} \|g\|_{L^2}, \quad \|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

For  $1 \leq p \leq \infty$ , Minkowski's integral inequality is

$$\left\| \int_Y F(\cdot, y) d\nu(y) \right\|_{L^p(X)} \leq \int_Y \|F(\cdot, y)\|_{L^p(X)} d\nu(y).$$

If  $\mu(X) = 1$ ,  $\Phi$  is convex, and  $f, \Phi \circ f \in L^1(X)$ , then

$$\Phi\left(\int_X f d\mu\right) \leq \int_X \Phi(f) d\mu.$$

If  $0 < \alpha < \beta \leq 1$  and  $\varepsilon > 0$ , then

$$[u]_{C^{0,\alpha}(\Omega)} \leq \varepsilon [u]_{C^{0,\beta}(\Omega)} + C(\varepsilon, \alpha, \beta, \Omega) \|u\|_{L^\infty(\Omega)}.$$

**Proposition A.1.2** (Young inequalities). *Let  $a, b \geq 0$ ,  $1 < p < \infty$ , and  $p' = p/(p-1)$ . Then*

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}, \quad ab \leq \varepsilon a^2 + \frac{b^2}{4\varepsilon} \quad (\varepsilon > 0).$$

If  $1 \leq p, q, r \leq \infty$  and  $1 + r^{-1} = p^{-1} + q^{-1}$ , then

$$f \in L^p(\mathbb{R}^n) \quad \wedge \quad g \in L^q(\mathbb{R}^n) \implies \|f * g\|_{L^r(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^q(\mathbb{R}^n)}.$$

**Proposition A.1.3** (Iteration and absorption). *Let  $A > 0$ ,  $B > 1$ ,  $\delta > 0$ , and  $Y_j \geq 0$ . Then*

$$Y_{j+1} \leq AB^j Y_j^{1+\delta} \quad \wedge \quad Y_0 \leq A^{-1/\delta} B^{-1/\delta^2} \implies Y_j \leq A^{-1/\delta} B^{-j/\delta - 1/\delta^2} \longrightarrow 0.$$

Let  $0 < \vartheta < 1$ ,  $\gamma > 0$ , and let  $\Phi : [r_0, R] \rightarrow [0, \infty)$  be bounded. Then

$$\Phi(r) \leq \vartheta \Phi(s) + A(s-r)^{-\gamma} \quad (r_0 \leq r < s \leq R) \implies \Phi(r) \leq C(\vartheta, \gamma) A(R-r)^{-\gamma}.$$

**Proposition A.1.4** (Markov and Chebyshev). *Let  $Y \geq 0$ ,  $Z \in L^2$ ,  $t > 0$ , and let  $\{A_j\}$  be events. Then*

$$\mathbb{P}\left(\bigcup_j A_j\right) \leq \sum_j \mathbb{P}(A_j), \quad \mathbb{P}(Y \geq t) \leq \frac{\mathbb{E}Y}{t}, \quad \mathbb{P}(|Z - \mathbb{E}Z| \geq t) \leq \frac{\text{Var}(Z)}{t^2}.$$

#### REFERENCES

- [1] E. Kim and Z. Tao, *Polynomial bounds for eigenfunctions and eigenvalues on random covers of hyperbolic surfaces*, preprint, 2026.
- [2] O. Savin, *Analysis II*. Lecture notes, MATH6152GR.